

True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q5/5

1601

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 5.NF.B.6

FORWARD 2

Surface area of prisms with fractional edge lengths • Grade 7 • Geometry •
7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q1/5

1603

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

59 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q1/5

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q2/5

1604

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

37 1/2 square inches

GRADING NOTE 37.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q2/5

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its surface area?

- A.** 30 square inches
- B.** 40 square inches
- C.** 67 square inches
- D.** $33\frac{1}{2}$ square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q3/5

1605

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 67 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q3/5

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q4/5

1606

QUESTION ID

S36-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**13 1/2 square
meters**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q4/5

True or false?

A prism that is 2 ft by $1\frac{1}{2}$ ft by 1 ft has a surface area of 13 square feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q5/5

1607

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q5/5

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{7} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q1/5

1610

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q1/5

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q2/5

1611

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q2/5

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q3/5

1612

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q3/5

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q4/5

1613

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q4/5

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q5/5

1614

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q1/5

1616

QUESTION ID

S37-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q1/5

Which unit fraction is the smallest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q2/5

1617

QUESTION ID

S37-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q2/5

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q3/5

1618

QUESTION ID

S37-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q3/5

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q4/5

1619

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q4/5

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q5/5

1620

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q1/5

1622

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q1/5

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q2/5

1623

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q2/5

Which expression shows 21 shared equally among 7 people?

- A. $21 \div 7$
- B. $21 - 7$
- C. $7 \div 21$
- D. 21×7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q3/5

1624

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $21 \div 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q3/5

Divide.

$$16 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q4/5

1625

QUESTION ID

S37-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q4/5

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q5/5

1626

QUESTION ID

S37-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{3} = \frac{?}{12}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q1/5

1628

QUESTION ID

S37-B4-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q1/5

True or false?

$$\frac{1}{2} = \frac{5}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q2/5

1629

QUESTION ID

S37-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q2/5

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{1}{8}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q3/5

1630

QUESTION ID

S37-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{1}{5} = \frac{3}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q4/5

1631

QUESTION ID

S37-B4-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q4/5

True or false?

$$\frac{1}{6} = \frac{3}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q5/5

1632

QUESTION ID

S37-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q5/5

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q1/5

1634

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q1/5

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q2/5

1635

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q2/5

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q3/5

1636

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q3/5

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q4/5

1637

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q4/5

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q5/5

1638

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q1/5

1640

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q1/5

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q2/5

1641

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q2/5

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q3/5

1642

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q3/5

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q4/5

1643

QUESTION ID

S37-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q4/5

True or false?

$$-(\frac{1}{2} \div 5) = (-\frac{1}{2}) \div 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q5/5

1644

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q5/5

Use models to divide a unit fraction by a whole number and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 69 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

The bar shows $\frac{1}{3}$ shaded.

Split every third into 2 equal parts to find $\frac{1}{3} \div 2$. Then write the quotient.



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q1/5

1647

QUESTION ID

S69-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Each of the 3 parts split into 2 equal parts (6 small parts in all); the shaded third is now 2 small parts

Written: $\frac{1}{6}$

GRADING NOTE Grade both: all 3 parts split in half, and the quotient $\frac{1}{6}$ (one small part is $\frac{1}{6}$ of the whole bar).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q1/5

Explain how you can use multiplication to check that

$$\frac{1}{4} \div 3 = \frac{1}{12}.$$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q2/5

1648

QUESTION ID

S69-M-Q2

TYPE

Short answer

CORRECT ANSWER

**Multiply the quotient by the divisor:
 $\frac{1}{12} \times 3 = \frac{3}{12} = \frac{1}{4}$,
which is the number
that was divided, so
the quotient is correct.**

GRADING NOTE Must multiply $\frac{1}{12}$ by 3 (or $3 \times \frac{1}{12}$) and get back $\frac{1}{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q2/5

Which equation checks $\frac{1}{5} \div 4 = \frac{1}{20}$?

A. $\frac{1}{20} \div 4 = \frac{1}{80}$

B. $\frac{1}{5} + 4 = 4\frac{1}{5}$

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

D. $\frac{1}{5} \times 4 = \frac{4}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q3/5

1649

QUESTION ID

S69-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q3/5

True or false?

$\frac{1}{2} \div 3 = \frac{1}{6}$ because 3 groups of $\frac{1}{6}$ make $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q4/5

1650

QUESTION ID

S69-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q4/5

4 friends share $\frac{1}{2}$ of a pizza equally.

How much of the whole pizza does each friend get?

Describe a model that shows it.

Share: _____

Model: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q5/5

1651

QUESTION ID

S69-M-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ of the pizza

GRADING NOTE Model: the half is split into 4 equal parts; the whole pizza then has 8 such parts, so each part is $\frac{1}{8}$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q5/5

SET 69 • 5.NF.B.7.a

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A bar is cut into 8 equal parts.

What fraction of the bar is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q1/5

1653

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q1/5

True or false?

2 parts of size $\frac{1}{6}$ make $\frac{2}{6}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q2/5

1654

QUESTION ID

S69-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q2/5

What fraction of the bar is shaded?



A. $\frac{5}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{6}$

D. $\frac{6}{1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q3/5

1655

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q3/5

A whole is cut into 3 equal parts.

What fraction of the whole is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q4/5

1656

QUESTION ID

S69-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{4}$ is one of 3 equal parts of a whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q5/5

1657

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • 5.NF.B.7.a

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{3}{12}$ as a whole number times $\frac{1}{12}$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q1/5

1659

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 × 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q1/5

True or false?

$$4 \times \frac{1}{8} = \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q2/5

1660

QUESTION ID

S69-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q2/5

Which fraction equals $6 \times \frac{1}{10}$?

A. $\frac{1}{60}$

B. $\frac{6}{10}$

C. $\frac{10}{6}$

D. $\frac{6}{60}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q3/5

1661

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q3/5

Multiply.

$$2 \times \frac{1}{6}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q4/5

1662

QUESTION ID

S69-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2/6

GRADING NOTE 1/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q4/5

True or false?

$$3 \times \frac{1}{9} = \frac{3}{27}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q5/5

1663

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q5/5

SET 69 • 5.NF.B.7.a

FORWARD 1

Divide fractions and check with multiplication • Grade 6 • The Number System
• 6.NS.A.1

Divide.

$$\frac{2}{3} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q1/5

1665

QUESTION ID

S69-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE 2/12 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q1/5

True or false?

$$\frac{3}{4} \div \frac{1}{4} = 3 \text{ because } 3 \times \frac{1}{4} = \frac{3}{4}.$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q2/5

1666

QUESTION ID

S69-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q2/5

Which equation checks $\frac{2}{3} \div \frac{3}{4} = \frac{8}{9}$?

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

B. $\frac{3}{4} \div \frac{8}{9} = \frac{27}{32}$

C. $\frac{8}{9} \times \frac{2}{3} = \frac{16}{27}$

D. $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q3/5

1667

QUESTION ID

S69-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q3/5

Divide.

$$\frac{4}{5} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q4/5

1668

QUESTION ID

S69-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2} \div \frac{1}{3} = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q5/5

1669

QUESTION ID

S69-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q5/5

SET 69 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{4} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q1/5

1671

QUESTION ID

S69-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q1/5

True or false?

$$-\frac{1}{3} \div 5 = -\frac{1}{15}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q2/5

1672

QUESTION ID

S69-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q2/5

Find the quotient.

$$\frac{1}{6} \div (-3)$$

A. $\frac{1}{18}$

B. $-\frac{1}{2}$

C. -18

D. $-\frac{1}{18}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q3/5

1673

QUESTION ID

S69-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $-\frac{1}{18}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q3/5

Find the quotient.

$$\left(-\frac{1}{2}\right) \div (-4)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q4/5

1674

QUESTION ID

S69-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q4/5

True or false?

$$\left(-\frac{1}{5}\right) \div 2 = \frac{1}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q5/5

1675

QUESTION ID

S69-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q5/5

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q1/5

1678

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q1/5

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q2/5

1679

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q2/5

How many $\frac{1}{3}$ s are in 7?

A. 10

B. 21

C. $\frac{7}{3}$

D. $\frac{1}{21}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q3/5

1680

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q3/5

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q4/5

1681

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q4/5

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q5/5

1682

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q5/5

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q1/5

1684

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q1/5

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q2/5

1685

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q2/5

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q3/5

1686

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $24 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q3/5

How many 4s are in 32?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q4/5

1687

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q4/5

A 30-inch ribbon is cut into 5-inch pieces.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q5/5

1688

QUESTION ID

S38-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q1/5

1690

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q1/5

True or false?

There are 3 thirds in 2 wholes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q2/5

1691

QUESTION ID

S38-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q2/5

How many $\frac{1}{6}$ s make 2 wholes?

- A.** 2
- B.** 8
- C.** 12
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q3/5

1692

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q3/5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q4/5

1693

QUESTION ID

S38-B2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

There are 2 halves in 2 wholes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q5/5

1694

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q1/5

1696

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q1/5

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q2/5

1697

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q2/5

How many $\frac{2}{3}$ -cup servings are in 8 cups?

- A. 6
- B. $\frac{1}{12}$
- C. $\frac{16}{3}$
- D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q3/5

1698

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q3/5

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q4/5

1699

QUESTION ID

S38-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q4/5

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q5/5

1700

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q1/5

1702

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q1/5

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q2/5

1703

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q2/5

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A. $\frac{1}{12}$ mile per hour
- B. 3 miles per hour
- C. 8 miles per hour
- D. 12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q3/5

1704

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q3/5

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q4/5

1705

QUESTION ID

S38-F2-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q5/5

1706

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q5/5

Use models to divide a whole number by a unit fraction and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.b
BACKWARD 1	Write whole numbers as sums of unit fractions	4.NF.B.3.b
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide whole numbers by fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 70 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Each bar is 1 whole.

Split each whole into fourths to find $2 \div \frac{1}{4}$. Then write the quotient.

1 whole

1 whole

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q1/5

1709

QUESTION ID

S70-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: Each of the
2 bars split into 4
equal parts (8
fourths in all)**

Written: 8

GRADING NOTE Grade both: both wholes split into fourths, and the quotient 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q1/5

Explain how to use multiplication to check that

$$5 \div \frac{1}{3} = 15.$$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q2/5

1710

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

**$15 \times 1/3 = 15/3 = 5$,
which is the number
that was divided, so
15 is correct.**

GRADING NOTE Must multiply 15 by $1/3$ (or $1/3 \times 15$) and get back 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q2/5

Which equation checks $6 \div \frac{1}{2} = 12$?

A. $12 \div \frac{1}{2} = 24$

B. $6 \times \frac{1}{2} = 3$

C. $12 \times \frac{1}{2} = 6$

D. $6 \times 12 = 72$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q3/5

1711

QUESTION ID

S70-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $12 \times (1/2) = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q3/5

True or false?

$3 \div \frac{1}{5} = 15$ because each of the 3 wholes has 5 fifths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q4/5

1712

QUESTION ID

S70-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q4/5

Write a story problem for $4 \div \frac{1}{2}$ and solve it. Describe a model that shows the answer.

Story and answer: _____

Model: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q5/5

1713

QUESTION ID

S70-M-Q5

TYPE

Short answer

CORRECT ANSWER

Answer 8. Example story: How many 1/2-cup scoops are in 4 cups?

GRADING NOTE The story must ask how many groups of $\frac{1}{2}$ are in 4 (a story about sharing 4 between 2 people gives 2 and is not correct). Model: 4 wholes, each split into 2 halves, make 8 halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q5/5

SET 70 • 5.NF.B.7.b

BACKWARD 1

Write whole numbers as sums of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write 1 as a sum of thirds.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q1/5

1715

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3} + \frac{1}{3} + \frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q1/5

True or false?

$$2 = \frac{5}{5} + \frac{5}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q2/5

1716

QUESTION ID

S70-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q2/5

Which sum equals 1?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

C. $\frac{1}{4} + \frac{1}{4}$

D. $\frac{1}{2} + \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q3/5

1717

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q3/5

Write 2 as a fraction with a denominator of 6.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q4/5

1718

QUESTION ID

S70-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

$$3 = \frac{3}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q5/5

1719

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • 5.NF.B.7.b

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{8}{4}$ as a whole number times $\frac{1}{4}$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q1/5

1721

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 × 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q1/5

True or false?

$$15 \times \frac{1}{3} = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q2/5

1722

QUESTION ID

S70-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q2/5

What is $10 \times \frac{1}{5}$?

A. 2

B. $\frac{1}{50}$

C. 50

D. $\frac{10}{50}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q3/5

1723

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q3/5

Multiply.

$$6 \times \frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q4/5

1724

QUESTION ID

S70-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 6/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q4/5

True or false?

$$12 \times \frac{1}{4} = 48$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q5/5

1725

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q5/5

SET 70 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions and check with multiplication • Grade 6 •
The Number System • 6.NS.A.1

Divide.

$$3 \div \frac{3}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q1/5

1727

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q1/5

True or false?

$$2 \div \frac{2}{3} = 3 \text{ because } 3 \times \frac{2}{3} = 2.$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q2/5

1728

QUESTION ID

S70-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q2/5

Which equation checks $4 \div \frac{2}{7} = 14$?

A. $4 \times 14 = 56$

B. $4 \times \frac{2}{7} = \frac{8}{7}$

C. $14 \div \frac{2}{7} = 49$

D. $14 \times \frac{2}{7} = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q3/5

1729

QUESTION ID

S70-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $14 \times (2/7) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q3/5

How many $\frac{3}{4}$ -foot pieces can be cut from a 9-foot board?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q4/5

1730

QUESTION ID

S70-F1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q4/5

True or false?

$$6 \div \frac{3}{4} = \frac{9}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q5/5

1731

QUESTION ID

S70-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q5/5

SET 70 • 5.NF.B.7.b

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-3 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q1/5

1733

QUESTION ID

S70-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q1/5

True or false?

$$4 \div \left(-\frac{1}{2}\right) = -8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q2/5

1734

QUESTION ID

S70-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q2/5

Find the quotient.

$$(-5) \div \left(-\frac{1}{3}\right)$$

A. 15

B. $\frac{5}{3}$

C. -15

D. $-\frac{5}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q3/5

1735

QUESTION ID

S70-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q3/5

Find the quotient.

$$-2 \div \frac{1}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q4/5

1736

QUESTION ID

S70-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q4/5

True or false?

$$(-6) \div \frac{1}{2} = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q5/5

1737

QUESTION ID

S70-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q5/5

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 3	Word problems: a whole number times a unit fraction	4.NF.B.4.c
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{4}$ -cup scoops of birdseed are in 3 cups?

Scoops: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q1/5

1740

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q1/5

4 students share $\frac{1}{3}$ liter of paint equally.

How much paint does each student get?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q2/5

1741

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{12}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q2/5

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q3/5

1742

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q3/5

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q4/5

1743

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q4/5

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q5/5

1744

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q5/5

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.

How many bracelets can she make?

Bracelets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q1/5

1746

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q1/5

True or false?

If 32 students sit 4 at each table, there are 6 tables.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q2/5

1747

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q2/5

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A. 8
- B. 200
- C. 35
- D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q3/5

1748

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q3/5

A 27-foot fence is split into 9 equal sections.

How long is each section?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q4/5

1749

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q4/5

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q5/5

1750

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 5.NF.B.7.c

BACKWARD 3

Word problems: a whole number times a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each scoop holds $\frac{1}{4}$ cup.

How many cups are in 8 scoops?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q1/5

1752

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2 cups

GRADING NOTE 8/4 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q1/5

True or false?

5 servings of $\frac{1}{3}$ pound each weigh $\frac{5}{3}$ pounds in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q2/5

1753

QUESTION ID

S39-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q2/5

Each piece of ribbon is $\frac{1}{2}$ meter long.

How long are 6 pieces in all?

- A.** $\frac{1}{12}$ meter **B.** $6\frac{1}{2}$ meters
C. 12 meters **D.** 3 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q3/5

1754

QUESTION ID

S39-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q3/5

Each lap is $\frac{1}{8}$ mile.

How far are 16 laps?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q4/5

1755

QUESTION ID

S39-B3-Q4

TYPE

Short answer

CORRECT ANSWER

2 miles

GRADING NOTE 16/8 miles is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q4/5

True or false?

4 cups that each hold $\frac{1}{3}$ liter hold $\frac{4}{12}$ liter in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q5/5

1756

QUESTION ID

S39-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q5/5

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q1/5

1758

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q1/5

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q2/5

1759

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q2/5

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q3/5

1760

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q3/5

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q4/5

1761

QUESTION ID

S39-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{4}{5}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q5/5

1762

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q1/5

1764

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

–36 feet per minute

GRADING NOTE "36 feet down per minute" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q1/5

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q2/5

1765

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-20°F per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q2/5

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

- A.** \$135 **B.** $-\$45$
C. $-\$15$ **D.** $-\$135$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q3/5

1766

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $-\$135$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q3/5

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q4/5

1767

QUESTION ID

S39-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q4/5

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q5/5

1768

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q5/5

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 4	Multiply whole numbers by 10, 100, and 1,000	4.NBT.A.1
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q1/5

1771

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q1/5

How many ounces are in 3 pounds?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q2/5

1772

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q2/5

A track is 2.5 kilometers long.

How many meters is that?

- A. 250 m
- B. 25,000 m
- C. 2,500 m
- D. 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q3/5

1773

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2,500 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q3/5

Convert.

6 yards = ____ feet

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q4/5

1774

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q4/5

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q5/5

1775

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q5/5

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q1/5

1777

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q1/5

True or false?

1 pound = 16 ounces

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q2/5

1778

QUESTION ID

S40-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q2/5

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q3/5

1779

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q3/5

How many meters are in 1 kilometer?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q4/5

1780

QUESTION ID

S40-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

1 yard = 12 inches

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q5/5

1781

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$16 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q1/5

1783

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q1/5

1783

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$1,000 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q2/5

1784

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q2/5

1784

True or false?

$$3 \times 36 = 98$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q3/5

1785

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q3/5

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q4/5

1786

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$60 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q5/5

1787

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q5/5

1787

SET 40 • 5.MD.A.1

BACKWARD 4

Multiply whole numbers by 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q1/5

1789

QUESTION ID

S40-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q1/5

1789

True or false?

$$7 \times 1,000 = 700$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q2/5

1790

QUESTION ID

S40-B4-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q2/5

$$42 \times 10 = ?$$

- A.** 4.2
- B.** 420
- C.** 4,200
- D.** 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q3/5

1791

QUESTION ID

S40-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 420

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$16 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q4/5

1792

QUESTION ID

S40-B4-Q4

TYPE

Short answer

CORRECT ANSWER

16,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q4/5

1792

True or false?

3×100 is 10 times as much as 3×10 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q5/5

1793

QUESTION ID

S40-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q5/5

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.
How many centimeters are in 5 inches?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q1/5

1795

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q1/5

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q2/5

1796

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q2/5

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q3/5

1797

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q3/5

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q4/5

1798

QUESTION ID

S40-F1-Q4

TYPE

Short answer

CORRECT ANSWER

33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q4/5

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q5/5

1799

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q5/5

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q1/5

1801

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/5 foot per minute

GRADING NOTE 0.2 foot per minute is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q1/5

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q2/5

1802

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q2/5

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q3/5

1803

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q3/5

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q4/5

1804

QUESTION ID

S40-F2-Q4

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q5/5

1805

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q5/5

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 4	Write tenths and hundredths as decimals	4.NF.C.6
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

30 inches = ____ feet

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q1/5

1808

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/2

GRADING NOTE 2.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q1/5

Convert.

750 milliliters = ____ liters

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q2/5

1809

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

GRADING NOTE 3/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q2/5

A rope is 45 feet long.

How many yards is that?

- A.** 15
- B.** 5
- C.** 135
- D.** 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q3/5

1810

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Convert.

40 ounces = ____ pounds

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q4/5

1811

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

2 1/2**GRADING NOTE** 2.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q4/5

1811

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q5/5

1812

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

1.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q5/5

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q1/5

1814

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q1/5

True or false?

1 kilogram = 1,000 grams

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q2/5

1815

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q2/5

How many feet are in 1 yard?

- A.** 10
- B.** 3
- C.** 36
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q3/5

1816

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q3/5

How many cups are in 1 pint?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q4/5

1817

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q4/5

True or false?

1 foot = 10 inches

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q5/5

1818

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$36 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q1/5

1820

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$64 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q2/5

1821

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q2/5

1821

True or false?

$$84 \div 7 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q3/5

1822

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q3/5

$$90 \div 3 = ?$$

- A.** 33
- B.** 300
- C.** 27
- D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q4/5

1823

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$5,000 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q5/5

1824

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q5/5

1824

SET 41 • 5.MD.A.1

BACKWARD 4

Write tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{75}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q1/5

1826

QUESTION ID

S41-B4-Q1

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q1/5

True or false?

$$\frac{5}{10} = 0.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q2/5

1827

QUESTION ID

S41-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q2/5

Which decimal is equal to $\frac{45}{100}$?

- A. 45
- B. 0.045
- C. 0.45
- D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q3/5

1828

QUESTION ID

S41-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q3/5

Write $2\frac{6}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q4/5

1829

QUESTION ID

S41-B4-Q4

TYPE

Short answer

CORRECT ANSWER

2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q4/5

True or false?

$$\frac{8}{10} = 0.08$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q5/5

1830

QUESTION ID

S41-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q5/5

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q1/5

1832

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.5

GRADING NOTE 8 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q1/5

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q2/5

1833

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q2/5

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q3/5

1834

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q3/5

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q4/5

1835

QUESTION ID

S41-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q4/5

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q5/5

1836

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q5/5

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q1/5

1838

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 2.54i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q1/5

Write an equation for the number of feet f in y yards.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q2/5

1839

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$f = 3y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q2/5

Which equation converts k kilograms to g grams?

- A. $k = 1,000g$
- B. $g = 1,000k$
- C. $g = k \div 1,000$
- D. $g = k + 1,000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q3/5

1840

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $g = 1,000k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q3/5

Write an equation for the number of pounds p in z ounces.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q4/5

1841

QUESTION ID

S41-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$p = z/16$

GRADING NOTE $p = z \div 16$ or $p = (1/16)z$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q4/5

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q5/5

1842

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q5/5

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.

How many feet of ribbon are left?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q1/5

1845

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q1/5

A recipe needs 2 pounds of flour. Jo has 20 ounces.

How many more ounces does she need?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q2/5

1846

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q2/5

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A.** 1,050 m
- B.** 2,050 m
- C.** 9,700 m
- D.** 862 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q3/5

1847

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2,050 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q3/5

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q4/5

1848

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q4/5

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q5/5

1849

QUESTION ID

S42-M-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q5/5

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.

What time does it end?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q1/5

1851

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3:55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q1/5

True or false?

3 quarters and 2 dimes make 85 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q2/5

1852

QUESTION ID

S42-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q2/5

Jake has 2 meters of string. He uses 75 cm.

How many centimeters are left?

- A. 1.25 cm
- B. 75 cm
- C. 125 cm
- D. 275 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q3/5

1853

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 125 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q3/5

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q4/5

1854

QUESTION ID

S42-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q4/5

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q5/5

1855

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1,600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q1/5

1857

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q1/5

True or false?

1 gallon = 4 quarts

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q2/5

1858

QUESTION ID

S42-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q2/5

How many centimeters are in 1 meter?

- A.** 1,000
- B.** 100
- C.** 10
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q3/5

1859

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q3/5

How many feet are in 3 yards?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q4/5

1860

QUESTION ID

S42-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q4/5

True or false?

1 liter = 100 milliliters

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q5/5

1861

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q1/5

1863

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 miles per quart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q1/5

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q2/5

1864

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q2/5

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

- A. \$72
- B. \$18
- C. \$24
- D. \$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q3/5

1865

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. \$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q3/5

A 24-ounce bag of rice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q4/5

1866

QUESTION ID

S42-F1-Q4

TYPE

Short answer

CORRECT ANSWER

\$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q5/5

1867

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q5/5

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q1/5

1869

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The 2-liter bottle

GRADING NOTE Bottle: \$0.90 per liter; can: \$1.20 per liter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q1/5

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q2/5

1870

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 minutes

GRADING NOTE 7.5 cm = 15 km; 15 km at 60 km per hour takes 1/4 hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q2/5

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

A. 1,050

B. 3

C. 4

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q3/5

1871

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q3/5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q4/5

1872

QUESTION ID

S42-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.5 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q4/5

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q5/5

1873

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q5/5

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare the difference in centers with the variability	7.SP.B.3

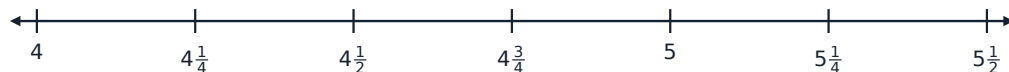
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q1/5

1876

QUESTION ID

S43-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 3 X's at 4
1/2, 1 X at 4 3/4, 2
X's at 5, 1 X at 5 1/4**

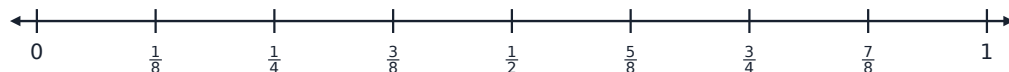
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q1/5

Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q2/5

1877

QUESTION ID

S43-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 1 X at $\frac{1}{8}$,
3 X's at $\frac{1}{4}$, 1 X at
 $\frac{3}{8}$, 2 X's at $\frac{1}{2}$**

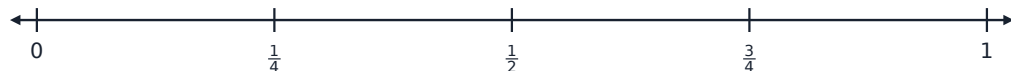
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q2/5

Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q3/5

1878

QUESTION ID

S43-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 1 X at $\frac{1}{4}$,
2 X's at $\frac{1}{2}$, 3 X's at
 $\frac{3}{4}$, 1 X at 1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q3/5

Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q4/5

1879

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

3

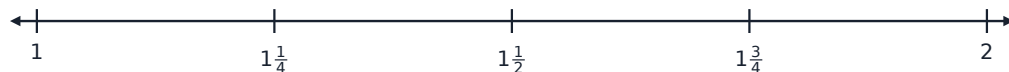
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q4/5

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q5/5

1880

QUESTION ID

S43-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 1 X at $1\frac{1}{4}$, 3 X's at $1\frac{1}{2}$, 1 X at $1\frac{3}{4}$, 2 X's at 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q5/5

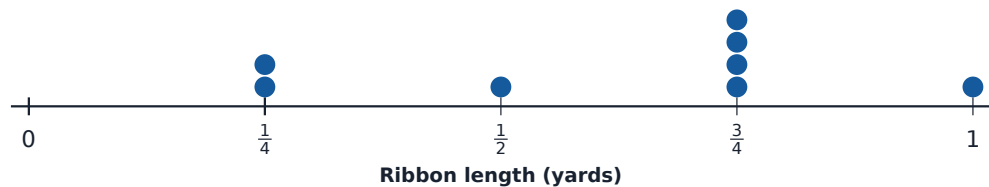
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q1/5

1882

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

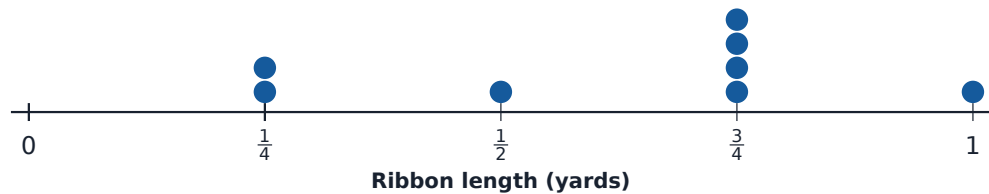
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

Two ribbons are $\frac{1}{4}$ yard long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q2/5

1883

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

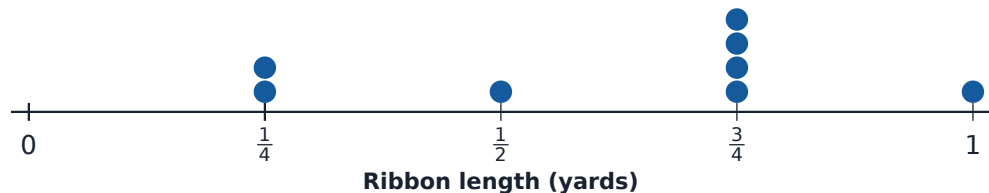
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q2/5

Which ribbon length is the most common?



- A. 1 yard B. $\frac{1}{2}$ yard C. $\frac{1}{4}$ yard D. $\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q3/5

1884

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

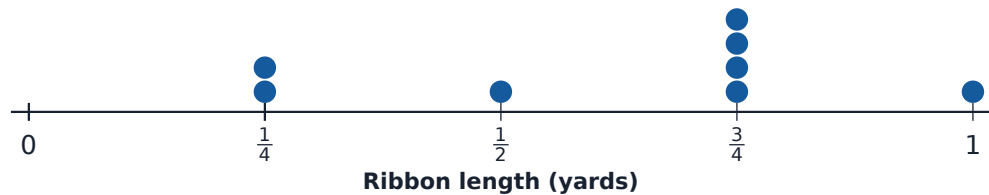
D. $\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q3/5

How many ribbons were measured?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q4/5

1885

QUESTION ID

S43-B1-Q4

TYPE

Short answer

CORRECT ANSWER

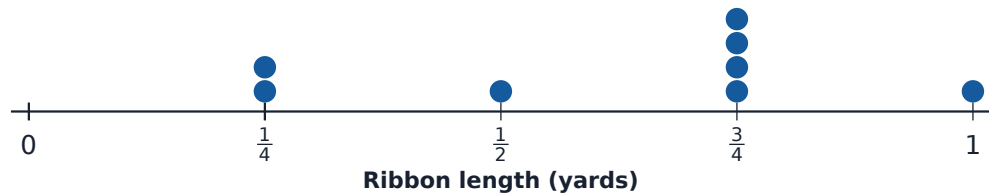
8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q4/5

What is the difference between the longest and the shortest ribbon?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q5/5

1886

QUESTION ID

S43-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

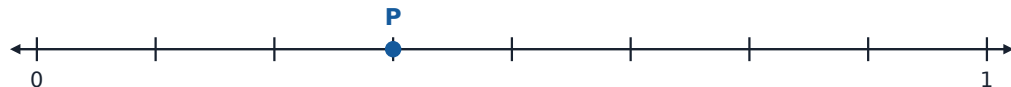
Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q1/5

1888

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q1/5

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q2/5

1889

QUESTION ID

S43-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q2/5

Into how many equal parts is the space from 0 to 1 divided?



A. 5

B. 7

C. 6

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q3/5

1890

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

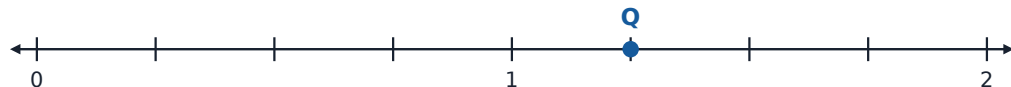
C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q3/5

What number is at point Q?



Q = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q4/5

1891

QUESTION ID

S43-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1 1/4

GRADING NOTE 5/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q4/5

True or false?

On a number line from 0 to 1 marked in eighths,
 $\frac{6}{8}$ is one mark to the left of 1.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q5/5

1892

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q1/5

1894

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q1/5

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q2/5

1895

QUESTION ID

S43-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q2/5

Which number line would you use for a line plot of 3, 4, 4, 7?

- A. A number line from 0 to 8
- B. A number line from 0 to 3
- C. A number line from 5 to 6
- D. A number line from 10 to 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q3/5

1896

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. A number line
from 0 to 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q3/5

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q4/5

1897

QUESTION ID

S43-B3-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 1 X at 2, 2 X's at 4, 1 X at 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q4/5

Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q5/5

1898

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

10 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q1/5

1900

QUESTION ID

S43-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 1 dot at 7,
2 dots at 8, 3 dots at
9, 1 dot at 10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q1/5

Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q2/5

1901

QUESTION ID

S43-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 2 dots at 0,
3 dots at 1, 2 dots at
2, 1 dot at 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q2/5

Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q3/5

1902

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**C. A number line
from 10 to 20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q3/5

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q4/5

1903

QUESTION ID

S43-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q4/5

True or false?

A dot plot shows every value in a data set.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q5/5

1904

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q5/5

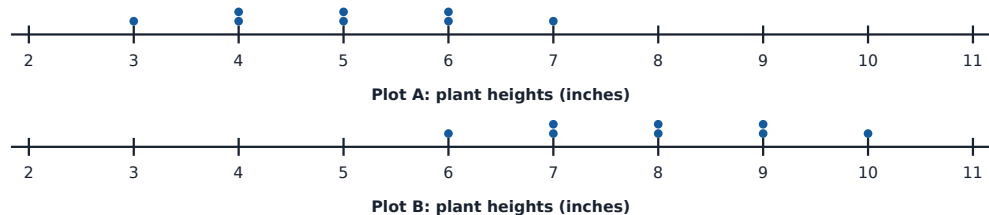
SET 43 • 5.MD.B.2

FORWARD 2

Compare the difference in centers with the variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Plot A has a mean of 5 inches and Plot B has a mean of 8 inches. Each plot has a mean absolute deviation (MAD) of 1 inch.

The difference in the means is how many times the MAD?



Times the MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q1/5

1906

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

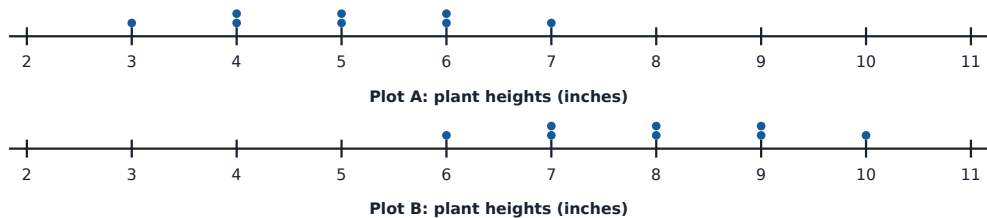
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q1/5

True or false?

The difference in the means (3 inches) is 3 times the MAD, so the two distributions overlap only a little.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q2/5

1907

QUESTION ID

S43-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q2/5

Which statement compares the two plots correctly?



- A. The centers differ by less than 1 MAD, so the plots are about the same.
- B. Plot A has much more variation than Plot B.
- C. The plots cannot be compared because they show different plants.
- D. The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q3/5

1908

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q3/5

Two teams have mean heights of 150 cm and 156 cm.
Each team has a MAD of 3 cm.

The difference in the means is how many times the
MAD?

Times the MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q4/5

1909

QUESTION ID

S43-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q4/5

True or false?

Two data sets have means of 20 and 21, and each has a MAD of 4. The difference in the means is large compared with the variability.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q5/5

1910

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q5/5

1910

Use operations on fractions to solve problems with line plot data

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 4	Rename halves and fourths as eighths	4.NF.A.1
BACKWARD 5	Add fractions with like denominators	4.NF.B.3.a
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.

What is the total amount of water?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q1/5

1913

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 $\frac{3}{8}$ liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q1/5

If all of the water were shared equally among the beakers, how much would each beaker hold?



Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q2/5

1914

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

19/64 liter

GRADING NOTE Total $2\frac{3}{8}$ liters $\div$ 8 beakers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q2/5

How much more water is in the fullest beaker than in the least full beaker?



- A. $\frac{1}{2}$ liter B. $\frac{1}{8}$ liter C. $\frac{5}{8}$ liter D. $\frac{3}{8}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q3/5

1915

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\frac{3}{8}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q3/5

What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q4/5

1916

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q4/5

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q5/5

1917

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

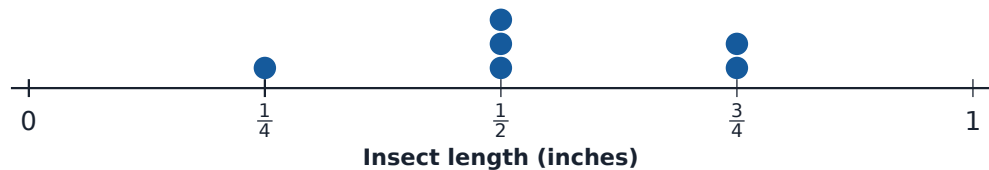
Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 1

Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q1/5

1919

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

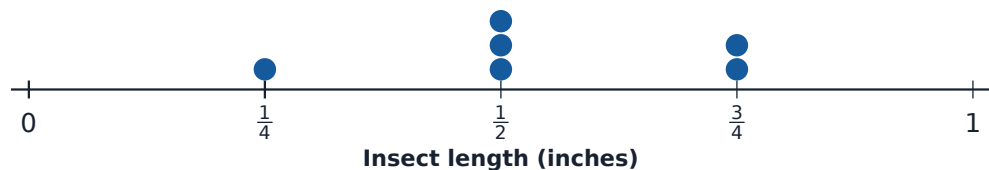
3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q1/5

What is the difference between the longest and the shortest insect?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q2/5

1920

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ inch

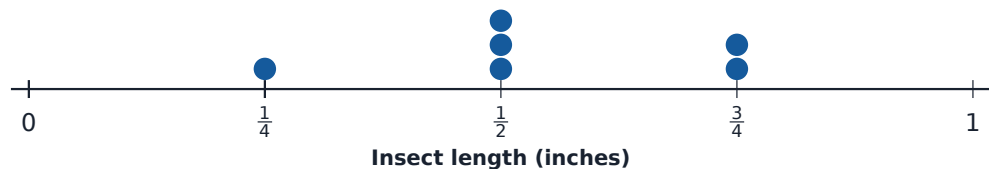
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q3/5

1921

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

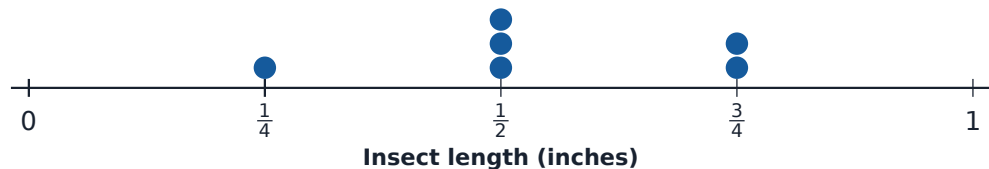
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q3/5

How many insects were measured?



A. 4

B. 3

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q4/5

1922

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

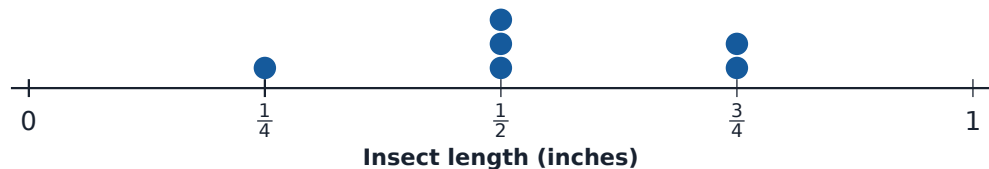
D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q4/5

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q5/5

1923

QUESTION ID

S44-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 1/2 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 4

Rename halves and fourths as eighths • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{1}{4}$ as eighths.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q1/5

1925

QUESTION ID

S44-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q1/5

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q2/5

1926

QUESTION ID

S44-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q2/5

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{7}{8}$

D. $\frac{4}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q3/5

1927

QUESTION ID

S44-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q4/5

1928

QUESTION ID

S44-B4-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q4/5

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q5/5

1929

QUESTION ID

S44-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 5

Add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{1}{8} + \frac{2}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q1/5

1931

QUESTION ID

S44-B5-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q1/5

True or false?

$$\frac{3}{8} + \frac{3}{8} = \frac{6}{16}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q2/5

1932

QUESTION ID

S44-B5-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q2/5

$$\frac{2}{8} + \frac{4}{8} + \frac{1}{8} = ?$$

A. $\frac{6}{8}$

B. $\frac{7}{8}$

C. $\frac{7}{24}$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q3/5

1933

QUESTION ID

S44-B5-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q3/5

Add.

$$\frac{5}{8} + \frac{3}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q4/5

1934

QUESTION ID

S44-B5-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE 8/8 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q4/5

True or false?

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q5/5

1935

QUESTION ID

S44-B5-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q5/5

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q1/5

1937

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q1/5

Find the mean.

12, 15, 21

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q2/5

1938

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q2/5

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q3/5

1939

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q3/5

The dot plot shows the goals scored in 6 games.

What is the mean number of goals?



Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q4/5

1940

QUESTION ID

S44-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 2 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q4/5

True or false?

The mean of 4, 4, and 10 is 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q5/5

1941

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q5/5

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q1/5

1943

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q1/5

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A. The populations are the same.
- B. Plants in the second population tend to be taller.
- C. Plants in the first population tend to be taller.
- D. No conclusion is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q2/5

1944

QUESTION ID

S44-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Plants in the second population tend to be taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q2/5

True or false?

Comparing the means of random samples is one way to compare two populations.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q3/5

1945

QUESTION ID

S44-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q3/5

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q4/5

1946

QUESTION ID

S44-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Class Y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q4/5

A random sample of 30 students from School A spends a mean of 40 minutes on homework. A random sample of 30 students from School B spends a mean of 50 minutes. The samples have similar variability.

What can you infer about all the students at the two schools?

Inference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q5/5

1947

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Students at School B probably tend to spend more time on homework than students at School A.

GRADING NOTE Must draw a conclusion about the populations (all students at each school), for example that School B students typically spend about 10 minutes more. Comparing only the two sample means, without a conclusion about the schools, is not enough.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q5/5

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q1/5

1950

QUESTION ID

S45-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q1/5

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q2/5

1951

QUESTION ID

S45-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 1 cubic unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q2/5

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q3/5

1952

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

1 cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q3/5

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q4/5

1953

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A sugar cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q4/5

True or false?

Volume measures the amount of space inside a solid figure.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q5/5

1954

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q5/5

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q1/5

1956

QUESTION ID

S45-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q1/5

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q2/5

1957

QUESTION ID

S45-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 1 square unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q2/5

How many unit squares cover a 3-by-3 square?

Unit squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q3/5

1958

QUESTION ID

S45-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q3/5

True or false?

Area is measured in cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q4/5

1959

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q4/5

A square has sides that are 1 inch long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q5/5

1960

QUESTION ID

S45-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 square inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q1/5

1962

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q1/5

True or false?

All the faces of a cube are squares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q2/5

1963

QUESTION ID

S45-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q2/5

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q3/5

1964

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q3/5

How many vertices (corners) does a cube have?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q4/5

1965

QUESTION ID

S45-B2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q4/5

True or false?

All the edges of a cube are the same length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q5/5

1966

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q1/5

1968

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q1/5

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q2/5

1969

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q2/5

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

- A. $\frac{1}{3}$ cubic foot B. $\frac{1}{9}$ cubic foot
C. $\frac{1}{27}$ cubic foot D. 1 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q3/5

1970

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/27 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q3/5

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q4/5

1971

QUESTION ID

S45-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q4/5

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q5/5

1972

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q1/5

1974

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q1/5

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q2/5

1975

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

27 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q2/5

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

- A.** 60 cubic inches
- B.** 30 cubic inches
- C.** 16 cubic inches
- D.** 600 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q3/5

1976

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q3/5

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q4/5

1977

QUESTION ID

S45-F2-Q4

TYPE

Short answer

CORRECT ANSWER

60 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q4/5

True or false?

Volume is measured in cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q5/5

1978

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q5/5

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q1/5

1981

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

30 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q1/5

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q2/5

1982

QUESTION ID

S46-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q2/5

A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units **B.** 18 cubic units
C. 54 cubic units **D.** 6 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q3/5

1983

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18 cubic units

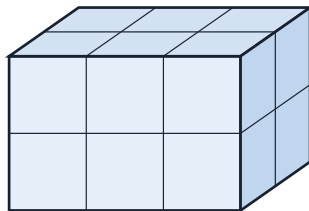
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q3/5

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q4/5

1984

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q4/5

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q5/5

1985

QUESTION ID

S46-M-Q5

TYPE

Short answer

CORRECT ANSWER

9 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q5/5

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q1/5

1987

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q1/5

True or false?

Overlapping squares give a wrong count for area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q2/5

1988

QUESTION ID

S46-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q2/5

A rectangle is covered by 20 unit squares.

What is its area?

- A. 10 square units
- B. 40 square units
- C. 20 cubic units
- D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q3/5

1989

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q3/5

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q4/5

1990

QUESTION ID

S46-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q4/5

True or false?

Area can be found by counting the unit squares that cover a shape.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q5/5

1991

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q1/5

1993

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q1/5

True or false?

3 rows of 5 make 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q2/5

1994

QUESTION ID

S46-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q2/5

Which equation matches an array with 2 rows of 5?

- A. $2 + 5 = 7$
- B. $5 + 5 = 10$
- C. $5 - 2 = 3$
- D. $2 + 2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q3/5

1995

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5 + 5 = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q3/5

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q4/5

1996

QUESTION ID

S46-B2-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q4/5

Write an addition equation for 3 rows of 4.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q5/5

1997

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q1/5

1999

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q1/5

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q2/5

2000

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q2/5

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

- A.** 27 cubic feet **B.** 9 cubic feet
C. 1 cubic foot **D.** 3 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q3/5

2001

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q3/5

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q4/5

2002

QUESTION ID

S46-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q4/5

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q5/5

2003

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6

A fish tank is 30 cm by 20 cm by 25 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q1/5

2005

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15,000 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q1/5

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q2/5

2006

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q2/5

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

- A.** 84 cubic meters
- B.** 42 cubic meters
- C.** 21 cubic meters
- D.** 14 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q3/5

2007

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 42 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q3/5

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q4/5

2008

QUESTION ID

S46-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.2 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q4/5

True or false?

Two prisms with the same base area and the same height have the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q5/5

2009

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q5/5

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

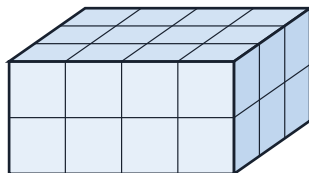
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q1/5

2012

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

24 cubic cm

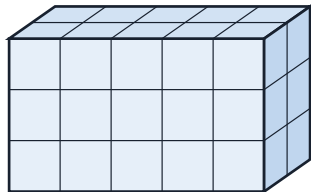
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q1/5

Each cube is 1 cubic inch.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q2/5

2013

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

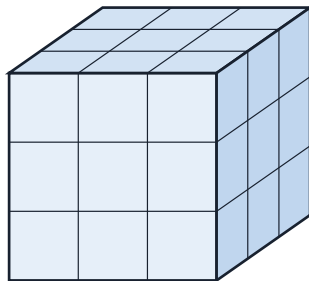
ID S47-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q2/5

Each cube is 1 cubic foot.

What is the volume of the prism?

- A.** 18 cubic feet **B.** 54 cubic feet
C. 27 cubic feet **D.** 9 cubic feet



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q3/5

2014

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 27 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q3/5

A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q4/5

2015

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

72 number cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

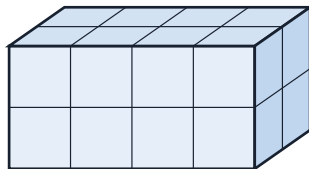
ID S47-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q4/5

True or false?

Each cube is 1 cubic cm.

The volume of the prism is
16 cubic cm.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q5/5

2016

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q5/5

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.

How many cubes are in the layer?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q1/5

2018

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q1/5

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q2/5

2019

QUESTION ID

S47-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q2/5

How many unit squares cover a 4-by-6 rectangle?

- A.** 10
- B.** 46
- C.** 24
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q3/5

2020

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q3/5

A rectangle is tiled with 7 rows of 3 unit squares.

How many squares are there?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q4/5

2021

QUESTION ID

S47-B1-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q4/5

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q5/5

2022

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.

How many cubes are there in all?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q1/5

2024

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q1/5

True or false?

4 layers of 10 cubes is 14 cubes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q2/5

2025

QUESTION ID

S47-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q2/5

Which expression shows 5 layers of 8 cubes?

- A.** $8 - 5$
- B.** 5×8
- C.** $5 + 8$
- D.** 58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q3/5

2026

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q4/5

2027

QUESTION ID

S47-B2-Q4

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q4/5

2027

There are 2 layers with 15 cubes in each layer.

How many cubes are there in all?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q5/5

2028

QUESTION ID

S47-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q1/5

2030

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q1/5

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q2/5

2031

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q2/5

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A. 4
- B. 8
- C. 16
- D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q3/5

2032

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

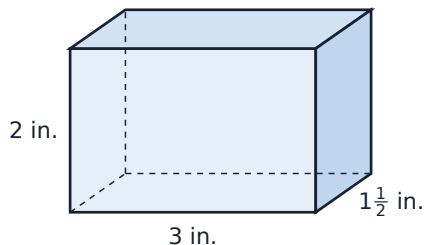
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q3/5

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q4/5

ID S47-F1-Q4

2033

QUESTION ID

S47-F1-Q4

TYPE

Short answer

CORRECT ANSWER

9 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q4/5

ID S47-F1-Q4

True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q5/5

2034

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q1/5

2036

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q1/5

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q2/5

2037

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q2/5

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet **B.** 30 cubic feet
C. 120 cubic feet **D.** 19 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q3/5

2038

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q3/5

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q4/5

2039

QUESTION ID

S47-F2-Q4

TYPE

Short answer

CORRECT ANSWER

108 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q4/5

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q5/5

2040

QUESTION ID

S47-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q5/5

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q1/5

2043

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

60 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q1/5

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q2/5

2044

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 12×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q2/5

True or false?

For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q3/5

2045

QUESTION ID

S48-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

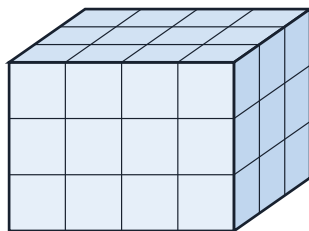
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q3/5

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q4/5

2046

QUESTION ID

S48-M-Q4

TYPE

Short answer

CORRECT ANSWER

36 cubic units

GRADING NOTE Base area $4 \times 3 = 12$; $12 \times 3 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q4/5

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q5/5

2047

QUESTION ID

S48-M-Q5

TYPE

Short answer

CORRECT ANSWER

72 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q5/5

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q1/5

2049

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q1/5

True or false?

The area of a 5-by-2 rectangle is 10 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q2/5

2050

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q2/5

Which expression gives the area of a 6-by-3 rectangle?

- A.** $6 - 3$
- B.** $6 + 3$
- C.** 6×3
- D.** $2 \times (6 + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q3/5

2051

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6×3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q3/5

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q4/5

2052

QUESTION ID

S48-B1-Q4

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q4/5

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q5/5

2053

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q1/5

2055

QUESTION ID

S48-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q1/5

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q2/5

2056

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q2/5

Which expression is equal to $4 \times 5 \times 2$?

- A. 9×2
- B. $20 + 2$
- C. $4 + 5 + 2$
- D. 4×10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q3/5

2057

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4×10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 2 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q4/5

2058

QUESTION ID

S48-B2-Q4

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q4/5

2058

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q5/5

2059

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q1/5

2061

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q1/5

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q2/5

2062

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q2/5

A prism is packed with cubes that have edges $\frac{1}{2}$ unit long: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

- A.** $6 + 4 + 2 = 12$ cubic units
- B.** $3 \times 2 \times 1 = 6$ cubic units
- C.** $6 \times 4 \times 2 = 48$ cubic units
- D.** $\frac{1}{2} \times 48 = 24$ cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q3/5

2063

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**B. $3 \times 2 \times 1 = 6$
cubic units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q3/5

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q4/5

2064

QUESTION ID

S48-F1-Q4

TYPE

Short answer

CORRECT ANSWER

27/64 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q4/5

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q5/5

2065

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q1/5

2067

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q1/5

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q2/5

2068

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

70 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q2/5

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

- A.** 1,008 m **B.** 7 m
C. 72 m **D.** 96 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q3/5

2069

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q3/5

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q4/5

2070

QUESTION ID

S48-F2-Q4

TYPE

Short answer

CORRECT ANSWER

100 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q4/5

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q5/5

2071

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q5/5

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Multiplying multi-digit numbers	4.NBT.B.5
BACKWARD 2	The area of the base	4.MD.A.3
FORWARD 1	Real-world volume with fractional edge lengths	6.G.A.2
FORWARD 2	Real-world volume problems with right prisms	7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q1/5

2074

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

4,320 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q1/5

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q2/5

2075

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

60 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q2/5

A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A.** 90 cubic cm
- B.** 36 cubic cm
- C.** 180 cubic cm
- D.** 18 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q3/5

2076

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 180 cubic cm

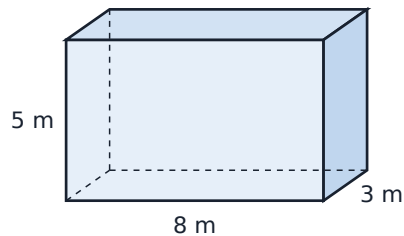
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q3/5

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q4/5

2077

QUESTION ID

S49-M-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q4/5

A cube-shaped box has edges that are 7 inches long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q5/5

2078

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

343 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q5/5

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$24 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q1/5

2080

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q1/5

2080

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$288 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q2/5

2081

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q2/5

2081

True or false?

$$36 \times 5 = 150$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q3/5

2082

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q3/5

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q4/5

2083

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 343

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$15 \times 24$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q5/5

2084

QUESTION ID

S49-B1-Q5

TYPE

Short answer

CORRECT ANSWER

360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q5/5

2084

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q1/5

2086

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

288 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q1/5

True or false?

A 9-by-4 rectangle has an area of 36 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q2/5

2087

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q2/5

The base of a box is 8 m by 3 m.

What is the area of the base?

- A. 11 square meters
- B. 24 square meters
- C. 22 square meters
- D. 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q3/5

2088

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q3/5

A square has sides that are 7 inches long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q4/5

2089

QUESTION ID

S49-B2-Q4

TYPE

Short answer

CORRECT ANSWER

49 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q4/5

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q5/5

2090

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q1/5

2092

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1 $\frac{1}{8}$ cubic feet

GRADING NOTE $\frac{9}{8}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q1/5

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q2/5

2093

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

37 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q2/5

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

- A.** $6\frac{1}{4}$ cubic cm **B.** 15 cubic cm
C. $7\frac{1}{2}$ cubic cm **D.** $15\frac{5}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q3/5

2094

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15 $\frac{5}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q3/5

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q4/5

2095

QUESTION ID

S49-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1 1/2 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q4/5

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q5/5

2096

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world volume problems with right prisms • Grade 7 • Geometry • 7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q1/5

2098

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q1/5

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q2/5

2099

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

75 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q2/5

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

- A.** 3 inches
- B.** 30 inches
- C.** 5 inches
- D.** 60 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q3/5

2100

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q3/5

A rectangular prism is 4.5 cm by 2 cm by 3 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q4/5

2101

QUESTION ID

S49-F2-Q4

TYPE

Short answer

CORRECT ANSWER

27 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q4/5

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q5/5

2102

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q5/5

Find a missing dimension when the volume is known

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Finding an unknown factor	3.OA.A.4
FORWARD 1	Solve one-step equations of the form $px = q$	6.EE.B.7
FORWARD 2	Solve two-step equations	7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q1/5

2105

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q1/5

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q2/5

2106

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

25 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q2/5

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A. 6 feet
- B. 30 feet
- C. 40 feet
- D. 227 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q3/5

2107

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q3/5

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q4/5

2108

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q4/5

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q5/5

2109

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q5/5

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q1/5

2111

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q1/5

Divide.

$$150 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q2/5

2112

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q2/5

True or false?

$$240 \div 8 = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q3/5

2113

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q3/5

$$84 \div 7 = ?$$

- A.** 12
- B.** 77
- C.** 11
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q4/5

2114

QUESTION ID

S50-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q4/5

Divide.

$$60 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q5/5

2115

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q1/5

2117

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q1/5

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q2/5

2118

QUESTION ID

S50-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q2/5

What number goes in the box?

$$4 \times \square = 36$$

- A. 9
- B. 8
- C. 32
- D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q3/5

2119

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q3/5

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q4/5

2120

QUESTION ID

S50-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q4/5

True or false?

If $8 \times \square = 56$, then $\square = 6$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q5/5

2121

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q1/5

2123

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q1/5

Solve for b.

$$6b = 150$$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q2/5

2124

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q2/5

Solve for w.

$$28w = 84$$

- A.** 56
- B.** 2,352
- C.** 112
- D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q3/5

2125

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q3/5

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q4/5

2126

QUESTION ID

S50-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$15h = 52.5; h = 3.5$
cm**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q4/5

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q5/5

2127

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q1/5

2129

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE $20(x + 2) = 120$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q1/5

Solve for x.

$$6(x + 3) = 54$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q2/5

2130

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q2/5

Solve for h.

$$12h + 8 = 68$$

- A.** 5
- B.** 6.3
- C.** 60
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q3/5

2131

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q3/5

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q4/5

2132

QUESTION ID

S50-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$12h + 40 = 100;$$

$$h = 5 \text{ feet}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q4/5

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q5/5

2133

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q5/5

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 3	The area formula for rectangles	4.MD.A.3
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q1/5

2136

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

112 cubic feet

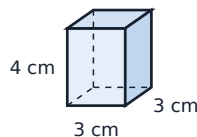
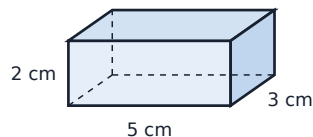
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q1/5

A figure is made of the two prisms shown, which do not overlap.

What is the total volume?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q2/5

2137

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

66 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q2/5

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units **B.** 8 cubic units
C. 10 cubic units **D.** 16 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q3/5

2138

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q3/5

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q4/5

2139

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

576 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q4/5

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q5/5

2140

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q5/5

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is split into two rectangles that do not overlap.
Their areas are 15 square units and 8 square units.
What is the area of the figure?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q1/5

2142

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

23 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q1/5

True or false?

The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q2/5

2143

QUESTION ID

S51-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

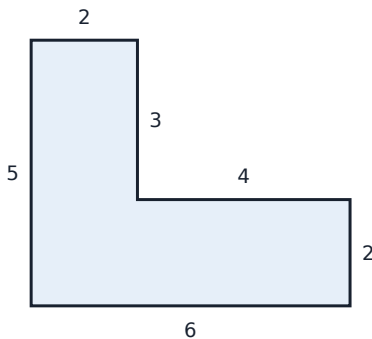
ID S51-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q2/5

The figure is split into a 6-by-2 rectangle along the bottom and a rectangle above it.

What is the area of the 6-by-2 rectangle?

- A.** 12 square units **B.** 16 square units
C. 6 square units **D.** 8 square units



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q3/5

2144

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q3/5

A figure is made of two squares that do not overlap.
Their areas are 9 square units and 25 square units.
To find the area of the figure, what do you do with the two areas?

- A.** Divide them
- B.** Multiply them
- C.** Subtract them
- D.** Add them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q4/5

2145

QUESTION ID

S51-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Add them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q4/5

A square with an area of 4 square units is cut out of a rectangle with an area of 48 square units.

What is the area of the part that is left?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q5/5

2146

QUESTION ID

S51-B1-Q5

TYPE

Short answer

CORRECT ANSWER

44 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 5.MD.C.5.c

BACKWARD 3

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 6 units by 4 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q1/5

2148

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A rectangle with an area of 32 square units and a length of 8 units has a width of 4 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q2/5

2149

QUESTION ID

S51-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q2/5

A rectangle is 10 m by 8 m.

What is its area?

- A.** 36 square meters
- B.** 800 square meters
- C.** 80 square meters
- D.** 18 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q3/5

2150

QUESTION ID

S51-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 80 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q3/5

A rectangle has an area of 45 square feet and a width of 5 feet.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q4/5

2151

QUESTION ID

S51-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q4/5

True or false?

A 4-by-8 rectangle has an area of 24 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q5/5

2152

QUESTION ID

S51-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q1/5

2154

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q1/5

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q2/5

2155

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q2/5

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

- A.** 2 cubic units **B.** $\frac{1}{2}$ cubic unit
- C.** 4 cubic units **D.** $1\frac{1}{2}$ cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q3/5

2156

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q3/5

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q4/5

2157

QUESTION ID

S51-F1-Q4

TYPE

Short answer

CORRECT ANSWER

27 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q4/5

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q5/5

2158

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q1/5

2160

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

96 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q1/5

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q2/5

2161

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3 cubic meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q2/5

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units **B.** 800 cubic units
C. 1,000 cubic units **D.** 40 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q3/5

2162

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 960 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q3/5

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q4/5

2163

QUESTION ID

S51-F2-Q4

TYPE

Short answer

CORRECT ANSWER

36 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q4/5

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q5/5

2164

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q5/5

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q1/5

2167

QUESTION ID

S52-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q1/5

In the ordered pair (7, 2), what does the 7 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q2/5

2168

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

**Start at the origin
and move 7 units
along the x-axis (to
the right).**

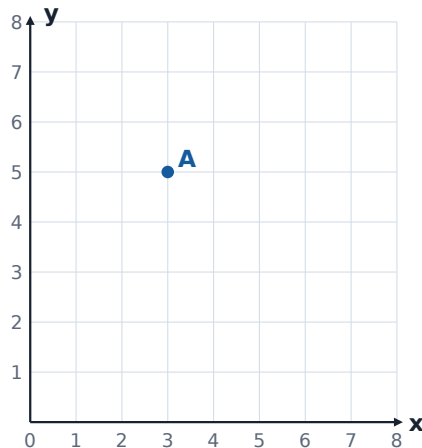
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q2/5

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q3/5

2169

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q3/5

True or false?

To reach (0, 6), you start at the origin and move 6 units to the right along the x-axis.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q4/5

2170

QUESTION ID

S52-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

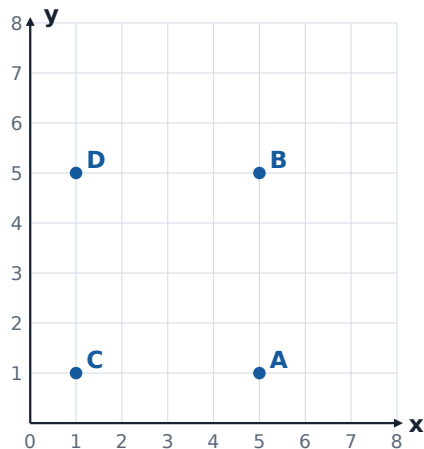
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q4/5

Which point is at (5, 1)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q5/5

2171

QUESTION ID

S52-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Point A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q5/5

SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q1/5

2173

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q1/5

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q2/5

2174

QUESTION ID

S52-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q2/5

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q3/5

2175

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q3/5

How many units is it from 0 to 9 on a number line?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q4/5

2176

QUESTION ID

S52-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q4/5

True or false?

On a number line, 3 is farther from 0 than 7 is.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q5/5

2177

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q1/5

2179

QUESTION ID

S52-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q1/5

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q2/5

2180

QUESTION ID

S52-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q2/5

How many degrees are in a right angle?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q3/5

2181

QUESTION ID

S52-B2-Q3

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q3/5

True or false?

Parallel lines never meet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q4/5

2182

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q4/5

Which letter is made of two perpendicular line segments?

- A. S
- B. L
- C. O
- D. C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q5/5

2183

QUESTION ID

S52-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q5/5

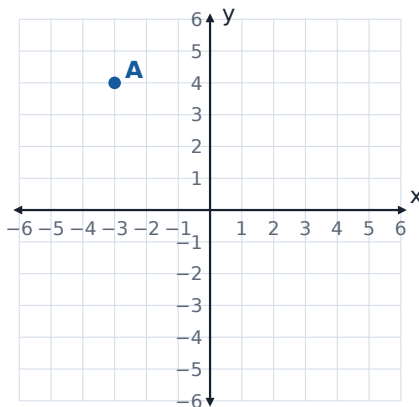
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q1/5

2185

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(-3, 4)

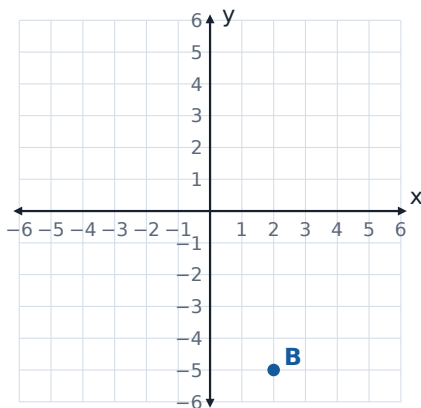
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q2/5

2186

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, -5)

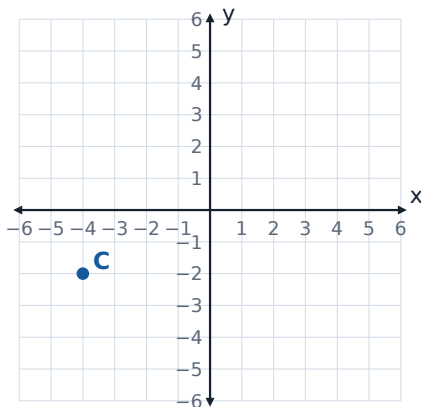
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q2/5

Which ordered pair names point C?

- A. $(-4, -2)$
- B. $(-2, -4)$
- C. $(4, 2)$
- D. $(-4, 2)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q3/5

2187

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(-4, -2)$

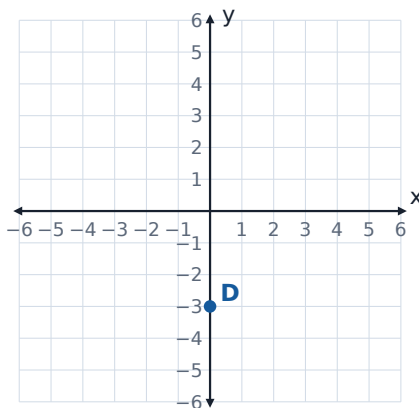
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q3/5

What are the coordinates of point D?

D = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q4/5

2188

QUESTION ID

S52-F1-Q4

TYPE

Short answer

CORRECT ANSWER

(0, -3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

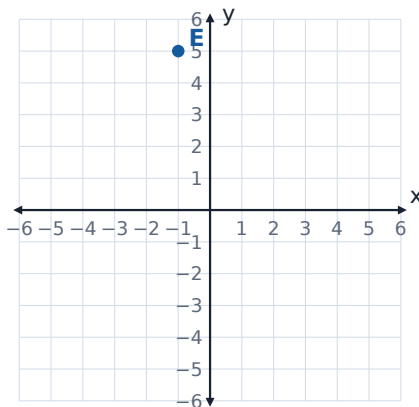
ID S52-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q4/5

True or false?

Point E is located at $(5, -1)$.

- A.** True
B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q5/5

2189

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q5/5

SET 52 • 5.G.A.1

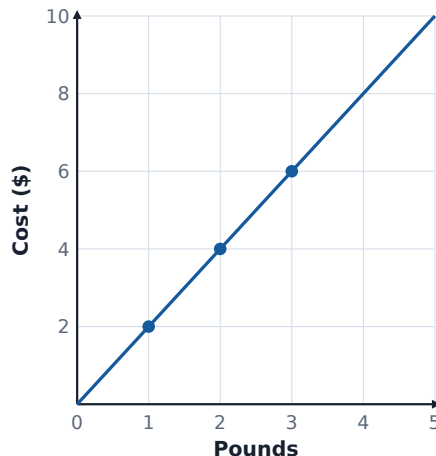
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q1/5

2191

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**3 pounds of pears
cost \$6.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

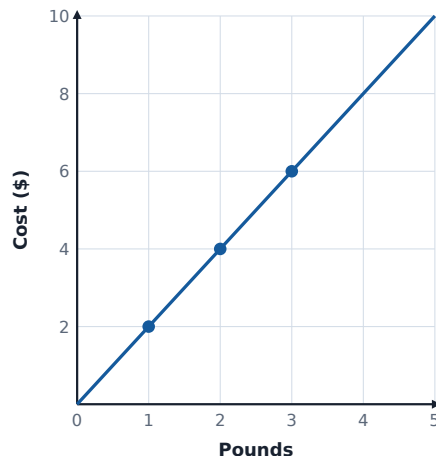
ID S52-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q1/5

The graph shows the cost of pears.

What does the point (1, 2) represent?

- A. The cost starts at \$2.
- B. 2 pounds cost \$1.
- C. \$1 buys 2 pounds.
- D. 1 pound costs \$2.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q2/5

2192

QUESTION ID

S52-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 1 pound costs \$2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q2/5

True or false?

On the graph of a proportional relationship between pounds and cost, the point (0, 0) means 0 pounds cost \$0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q3/5

2193

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q3/5

The graph of a proportional relationship passes through (1, r) and (4, 20).

What is r ?

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q4/5

2194

QUESTION ID

S52-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q4/5

A graph shows a proportional relationship between the time x , in hours, that a bus drives and the distance y , in miles. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q5/5

2195

QUESTION ID

S52-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. In 2 hours, the distance is 90 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q5/5

Graph points in the first quadrant

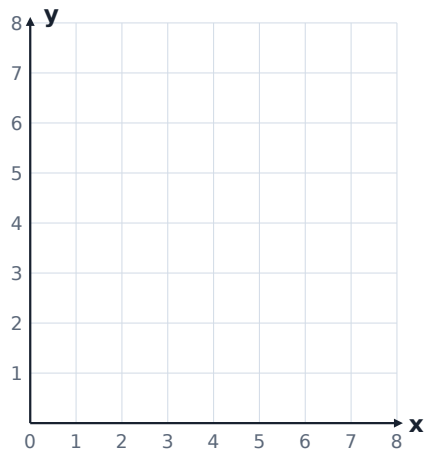
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Read a scaled bar graph	3.MD.B.3
BACKWARD 3	Fractions and whole numbers on a number line	3.NF.A.2
FORWARD 1	Reflect points across the axes	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points A(2, 7), B(6, 3), and C(0, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

2198

QUESTION ID

S53-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

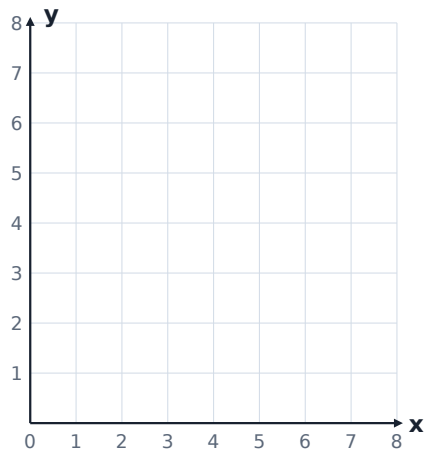
A plotted at (2, 7), B at (6, 3), and C at (0, 5) on the y-axis, each labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q2/5

2199

QUESTION ID

S53-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

D plotted at (4, 4), E at (7, 0) on the x-axis, and F at (1, 6), each labeled

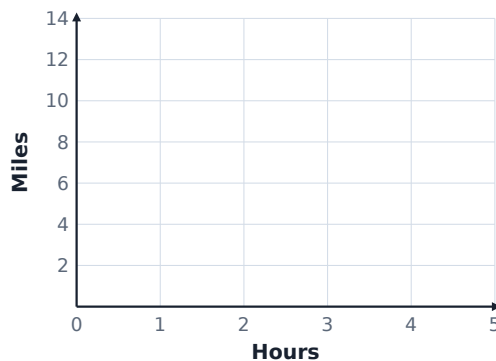
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q2/5

Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q3/5

2200

QUESTION ID

S53-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

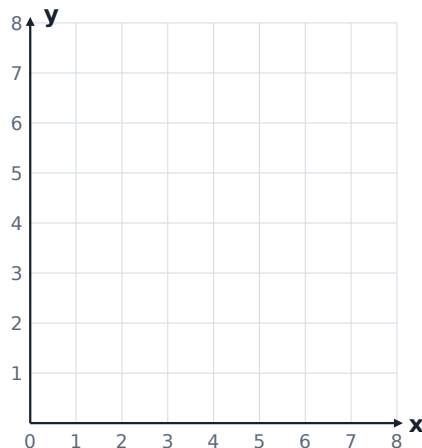
Points plotted at (1, 3), (2, 6), (3, 9), (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q3/5

Plot and label the points G(3, $2\frac{1}{2}$) and H($5\frac{1}{2}$, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

2201

QUESTION ID

S53-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

G plotted at (3, $2\frac{1}{2}$), halfway between 2 and 3 on the y-axis scale; H plotted at ($5\frac{1}{2}$, 4); each labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

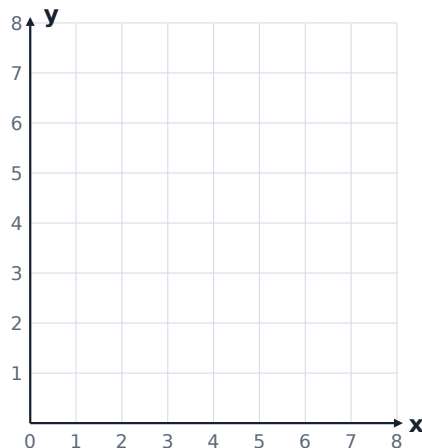
ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

Plot P(2, 3) and Q(2, 7).

How far apart are the two points?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q5/5

ID S53-M-Q5

2202

QUESTION ID

S53-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: P plotted at (2, 3) and Q plotted at (2, 7)

Written: 4 units

GRADING NOTE Grade both: both points at the correct places, and the distance 4 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q5/5

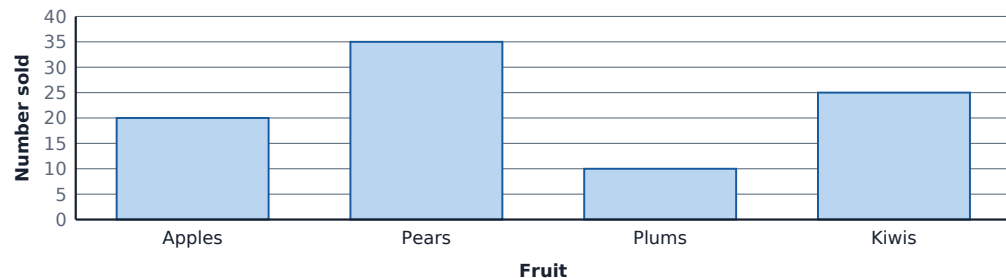
ID S53-M-Q5

SET 53 • 5.G.A.2

BACKWARD 2

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

What does each grid line on the scale stand for?



- A.** 10 **B.** 20 **C.** 5 **D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q1/5

2204

QUESTION ID

S53-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

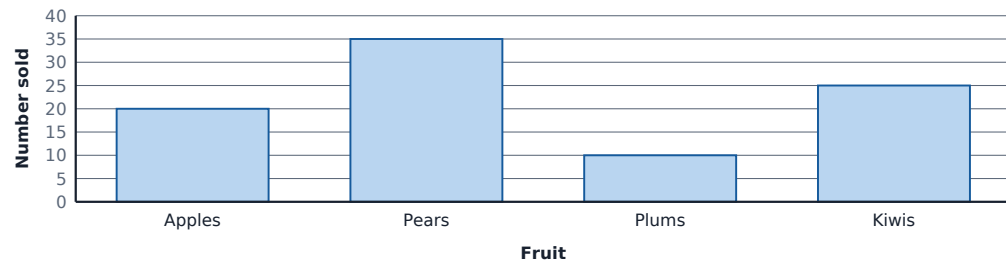
C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q1/5

How many pears were sold?



Pears: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q2/5

2205

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

35

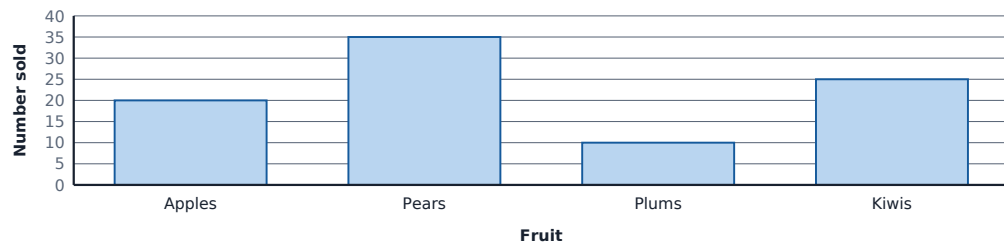
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

More kiwis than apples were sold.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q3/5

2206

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

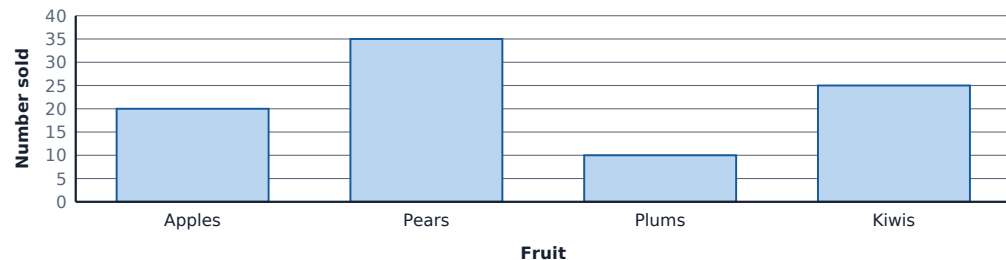
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q3/5

How many fewer plums than pears were sold?



Plums: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q4/5

2207

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

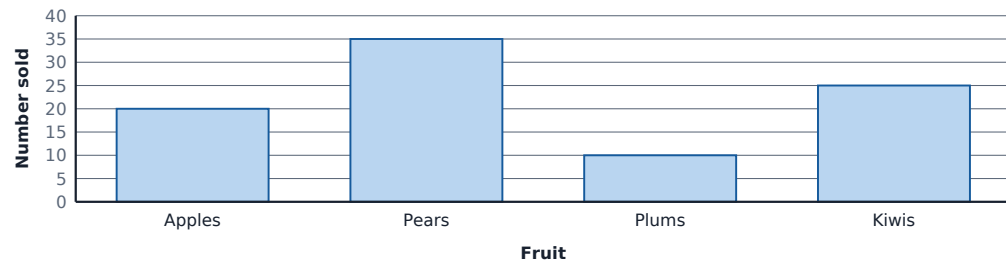
25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q4/5

How many apples and kiwis were sold in all?



Fruit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q5/5

2208

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

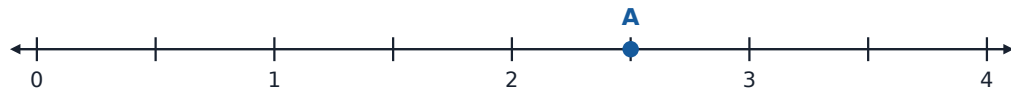
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 5.G.A.2

BACKWARD 3

Fractions and whole numbers on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q1/5

2210

QUESTION ID

S53-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/2

GRADING NOTE 5/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q1/5

True or false?

$2\frac{1}{2}$ is halfway between 2 and 3 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q2/5

2211

QUESTION ID

S53-B3-Q2

TYPE

True or false

CORRECT ANSWER

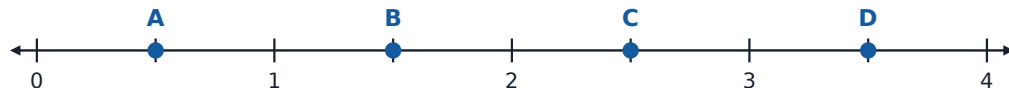
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q2/5

Which point is at $1\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q3/5

2212

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q4/5

2213

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q4/5

True or false?

On a number line, $5\frac{1}{2}$ is between 6 and 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q5/5

2214

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

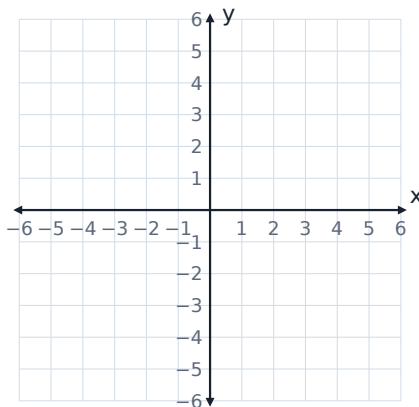
Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • 5.G.A.2

FORWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Plot and label A(2, 4). Then plot its reflection across the x-axis and label it B.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q1/5

2216

QUESTION ID

S53-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**A plotted at (2, 4)
and B at (2, -4),
each labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q1/5

The point (5, 2) is reflected across the x-axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q2/5

2217

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, -2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q2/5

The point $(-3, 5)$ is reflected across the x-axis.

What are the coordinates of its image?

- A. $(3, -5)$
- B. $(3, 5)$
- C. $(-3, -5)$
- D. $(5, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q3/5

2218

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(-3, -5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q3/5

The point $(-4, 1)$ is reflected across the y -axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q4/5

2219

QUESTION ID

S53-F1-Q4

TYPE

Short answer

CORRECT ANSWER

(4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q4/5

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q5/5

2220

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q5/5

SET 53 • 5.G.A.2

FORWARD 2

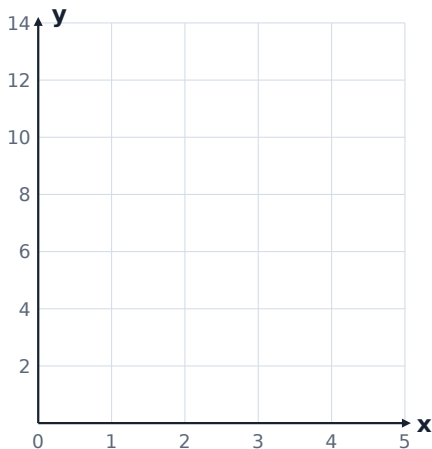
Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6), and (4, 12).

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q1/5

2222

QUESTION ID

S53-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at (1, 3), (2, 6) and (4, 12)

Written: Yes; the points lie on a straight line through (0, 0)

GRADING NOTE Grade both: all three points at the correct places, and "yes" with the reason (a straight line through the origin, or $y = 3x$ for every point).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

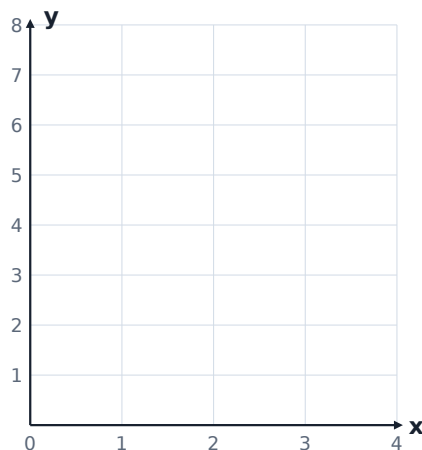
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q1/5

Graph the pairs $(0, 2)$, $(1, 4)$, and $(2, 6)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q2/5

2223

QUESTION ID

S53-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at $(0, 2)$, $(1, 4)$ and $(2, 6)$

Written: No; the points are on a line, but the line does not pass through $(0, 0)$

GRADING NOTE Grade both: all three points at the correct places, and "no" with the reason (the line does not go through the origin; when x is 0, y is 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q2/5

Which graph shows a proportional relationship?

- A. A curve through (0, 0)
- B. Points that are not on a line
- C. A straight line through (0, 3)
- D. A straight line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q3/5

2224

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. A straight line
through (0, 0)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The points (2, 1), (4, 2), and (6, 3) show a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q4/5

2225

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q4/5

Graph the pairs from the table.

Is the relationship proportional? Explain using your graph.

x	1	2	3
y	4	8	12



Answer: _____

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q5/5

2226

QUESTION ID

S53-F2-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at (1, 4), (2, 8) and (3, 12)

Written: Yes; the points lie on a straight line through (0, 0)

GRADING NOTE Grade both: all three points at the correct places, and "yes" with the reason (a straight line through the origin, or $y = 4x$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q5/5

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
BACKWARD 3	Solve problems with a scaled bar graph	3.MD.B.3
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

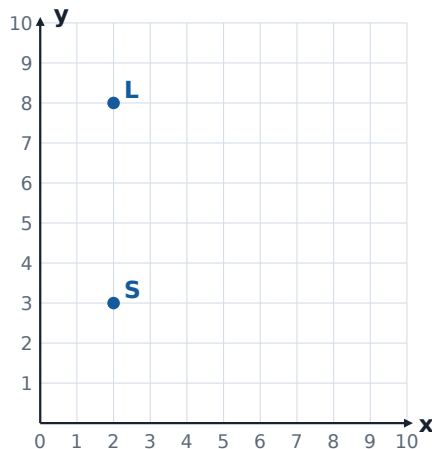
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q1/5

2229

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

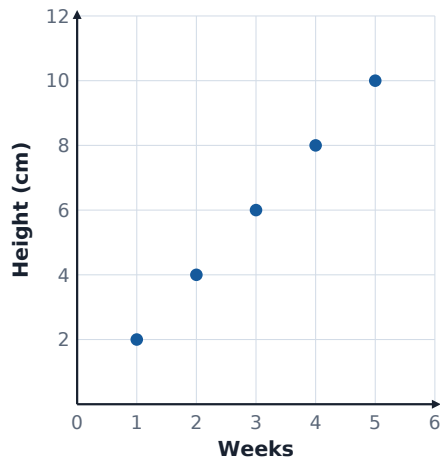
ID S54-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q1/5

The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q2/5

2230

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q2/5

The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

What does the point mean?

- A.** Working 3 hours earns \$12. **B.** Working 12 hours earns \$3.
C. The pay starts at \$12. **D.** Each hour earns \$3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q3/5

2231

QUESTION ID

S54-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. Working 3 hours
earns \$12.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

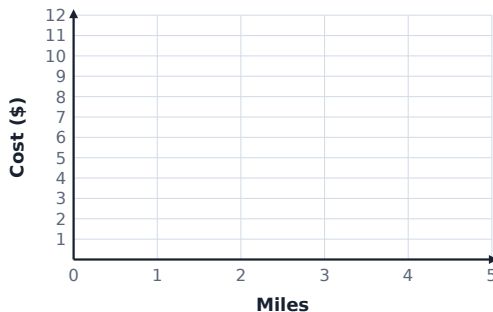
ID S54-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q3/5

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q4/5

2232

QUESTION ID

S54-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 5), (2, 7), (3, 9), (4, 11)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q4/5

On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q5/5

2233

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

600 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q5/5

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q1/5

2235

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q1/5

True or false?

The distance from 3 to 10 is 13 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q2/5

2236

QUESTION ID

S54-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q2/5

How many units apart are 1 and 8 on a number line?

- A.** 9
- B.** 8
- C.** 1
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q3/5

2237

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q3/5

What is the distance from 6 to 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q4/5

2238

QUESTION ID

S54-B2-Q4

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q4/5

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q5/5

2239

QUESTION ID

S54-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

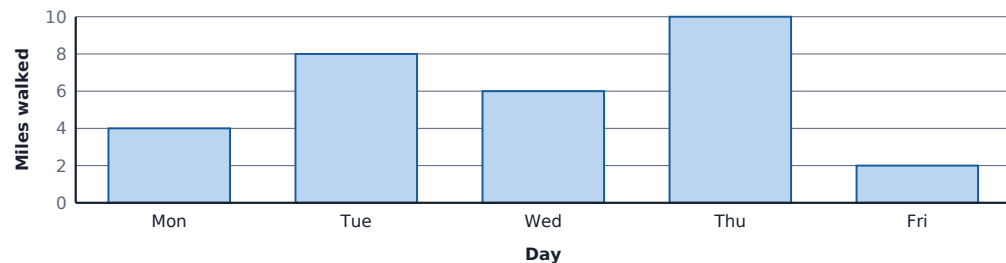
Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 5.G.A.2

BACKWARD 3

Solve problems with a scaled bar graph • Grade 3 • Measurement & Data •
3.MD.B.3

How many miles were walked on Thursday?



Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q1/5

2241

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

10

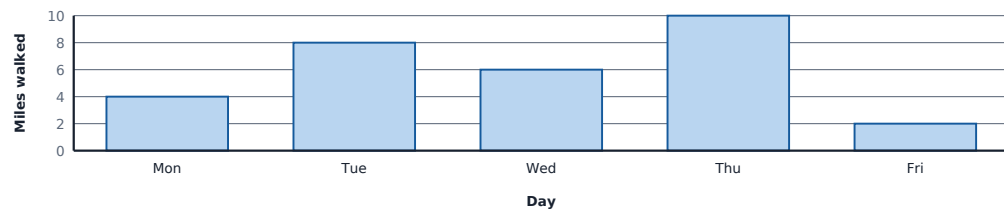
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q1/5

True or false?

The same number of miles were walked on Monday and Wednesday.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q2/5

2242

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

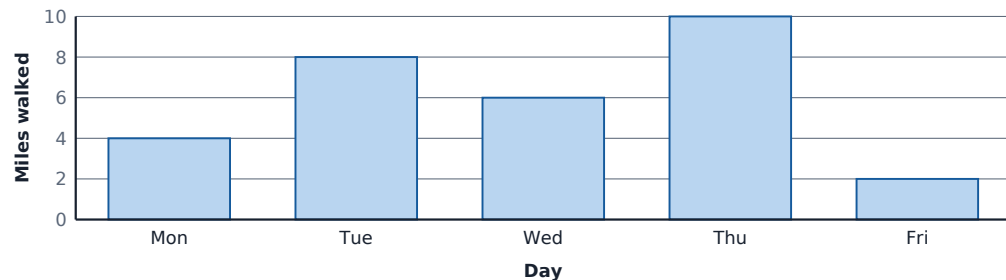
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q2/5

On which day were 8 miles walked?



A. Thursday B. Friday C. Monday D. Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

2243

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

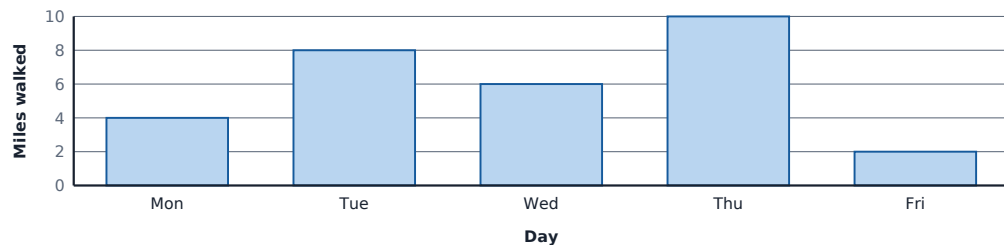
D. Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

How many more miles were walked on Thursday than on Friday?



Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q4/5

2244

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

8

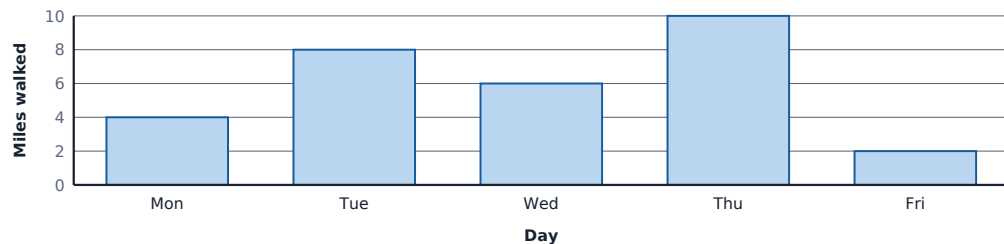
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q4/5

True or false?

Each grid line on the scale stands for 2 miles.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q5/5

2245

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q1/5

2247

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q1/5

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q2/5

2248

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q2/5

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

- A. 7 blocks
- B. 1 block
- C. 4 blocks
- D. 6 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q3/5

2249

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q3/5

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q4/5

2250

QUESTION ID

S54-F1-Q4

TYPE

Short answer

CORRECT ANSWER

16 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q4/5

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q5/5

2251

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q1/5

2253

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q1/5

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q2/5

2254

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q2/5

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

- A.** 12 m **B.** 35 m
C. 2 m **D.** 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q3/5

2255

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q3/5

A garden plan has corners at (0, 0), (6, 0), (6, 4), and (0, 4). Each unit is 3 feet.

What are the real length and width?

Length and width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q4/5

2256

QUESTION ID

S54-F2-Q4

TYPE

Short answer

CORRECT ANSWER

18 feet by 12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q5/5

2257

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q5/5

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Draw shapes with given conditions	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

Every parallelogram has opposite sides of equal length. A rhombus is a parallelogram, so every rhombus has opposite sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q1/5

2260

QUESTION ID

S55-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q1/5

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A.** Kite
- B.** Pentagon
- C.** Triangle
- D.** Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q2/5

2261

QUESTION ID

S55-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q2/5

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q3/5

ID S55-M-Q3

2262

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE A square is a quadrilateral with 4 sides of equal length, so it has every attribute of a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q3/5

ID S55-M-Q3

True or false?

All quadrilaterals have 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q4/5

2263

QUESTION ID

S55-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q4/5

Which statement is true?

- A. Every parallelogram is a square.
- B. Every square is a rectangle.
- C. Every quadrilateral is a rectangle.
- D. Every rectangle is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q5/5

2264

QUESTION ID

S55-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Every square is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q5/5

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q1/5

2266

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q1/5

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q2/5

2267

QUESTION ID

S55-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q2/5

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q3/5

2268

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q3/5

True or false?

A square has 4 sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q4/5

2269

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q4/5

Name a quadrilateral that always has 4 right angles.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q5/5

2270

QUESTION ID

S55-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Rectangle

GRADING NOTE Square is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q1/5

2272

QUESTION ID

S55-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q1/5

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q2/5

2273

QUESTION ID

S55-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q2/5

What do we call two lines that meet to form right angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q3/5

2274

QUESTION ID

S55-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Perpendicular lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q3/5

True or false?

An obtuse angle is smaller than a right angle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q4/5

2275

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q4/5

How many pairs of parallel sides does a rectangle have?

- A.** 4
- B.** 0
- C.** 2
- D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q5/5

2276

QUESTION ID

S55-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q1/5

2278

QUESTION ID

S55-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q1/5

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q2/5

2279

QUESTION ID

S55-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. A right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q2/5

How many right angles does a square have?

Right angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q3/5

2280

QUESTION ID

S55-B3-Q3

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q3/5

True or false?

A rhombus always has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q4/5

2281

QUESTION ID

S55-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q4/5

Which shape always has two pairs of parallel sides?

- A.** Rectangle
- B.** Kite
- C.** Triangle
- D.** Pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q5/5

2282

QUESTION ID

S55-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • 5.G.B.3

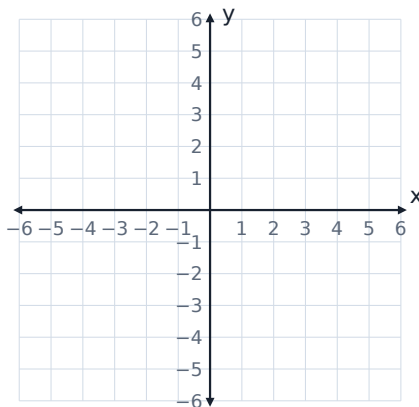
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot A(−2, 1), B(3, 1), C(3, −3), and D(−2, −3). Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q1/5

2284

QUESTION ID

S55-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A, B, C and D plotted and joined in order A-B-C-D-A

Written: Rectangle

GRADING NOTE Grade both: the four points and the four connecting segments, and the name. The sides are 5 and 4 units with four right angles, so "rectangle" is the most specific name. "Parallelogram" or "quadrilateral" are true but not the most specific; "square" is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

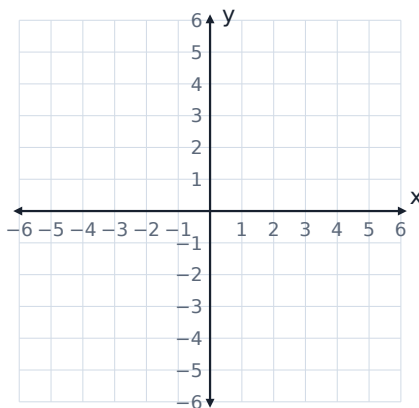
ID S55-F1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q1/5

Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q2/5

2285

QUESTION ID

S55-F1-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: E, F, G and H plotted and joined in order E-F-G-H-E

Written: Square

GRADING NOTE Grade both: the four points and the four connecting segments, and the name. All four sides are equal and all four angles are right angles, so "square" is the most specific name. "Rhombus", "rectangle", "parallelogram" and "quadrilateral" are true but not the most specific.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q2/5

The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

- A.** $(1, -1)$ **B.** $(-1, -1)$
C. $(-1, 4)$ **D.** $(4, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q3/5

2286

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(-1, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q3/5

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q4/5

2287

QUESTION ID

S55-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q4/5

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q5/5

2288

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • 5.G.B.3

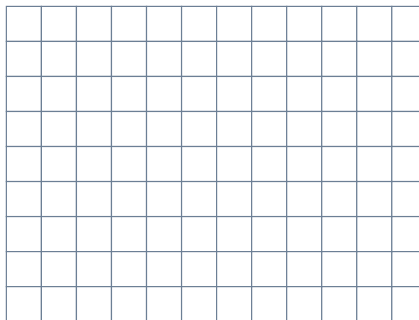
FORWARD 2

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a triangle with sides of 3 units, 4 units, and 5 units. Put the 3-unit and 4-unit sides along grid lines.

Classify the triangle by its angles.

Kind of triangle: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q1/5

ID S55-F2-Q1

2290

QUESTION ID

S55-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A triangle with a 3-unit side and a 4-unit side along grid lines meeting at a right angle, and the third side (5 units) joining their ends

Written: Right triangle

GRADING NOTE Grade both: the drawn side lengths 3 and 4 along grid lines with the third side joining them, and "right triangle".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q1/5

ID S55-F2-Q1

Can a triangle have sides of 2 cm, 2 cm, and 5 cm?
Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q2/5

2291

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE $2 + 2 = 4$ is less than 5, so the two short sides cannot meet to close the triangle.

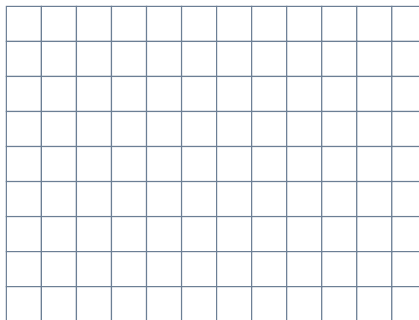
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q2/5

Draw a triangle with angles of 90° , 45° , and 45° . Make its two shorter sides 4 units long, along grid lines.

Could a triangle with these same angles be drawn in a different size?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q3/5

2292

QUESTION ID

S55-F2-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A right triangle with two 4-unit legs along grid lines meeting at a right angle, and the third side joining their ends

Written: Yes

GRADING NOTE Grade both: the drawing (two 4-unit legs at a right angle) and "yes": three angles do not fix the size, so larger or smaller triangles with the same angles can be drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q3/5

Can a triangle have angles of 100° , 50° , and 40° ?
Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q4/5

2293

QUESTION ID

S55-F2-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE The angles of a triangle add to 180° , but $100 + 50 + 40 = 190$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

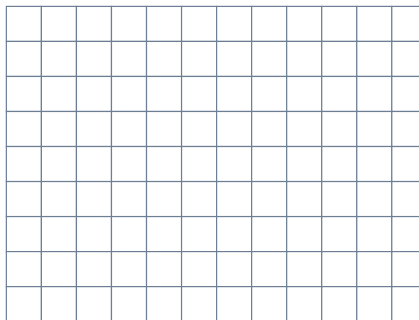
ID S55-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q4/5

Draw a quadrilateral that has exactly one pair of parallel sides.

What is the name of this kind of quadrilateral?

Name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q5/5

2294

QUESTION ID

S55-F2-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A quadrilateral with one pair of parallel sides (for example a horizontal top and bottom of different lengths) and one pair of sides that are not parallel

Written: Trapezoid

GRADING NOTE Grade both: exactly one pair of parallel sides in the drawing, and the name "trapezoid".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q5/5

Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Draw quadrilaterals and triangles with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A. Square, rectangle, parallelogram, quadrilateral
- B. Rectangle, square, quadrilateral, parallelogram
- C. Quadrilateral, parallelogram, rectangle, square
- D. Parallelogram, quadrilateral, square, rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q1/5

2297

QUESTION ID

S56-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**C. Quadrilateral,
parallelogram,
rectangle, square**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q1/5

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q2/5

2298

QUESTION ID

S56-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q2/5

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q3/5

2299

QUESTION ID

S56-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Quadrilateral and
parallelogram**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q3/5

A shape has 4 sides of equal length and 4 right angles.

What is the most specific name for it?

- A. Rectangle
- B. Rhombus
- C. Square
- D. Quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q4/5

2300

QUESTION ID

S56-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q4/5

True or false?

Every parallelogram is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q5/5

2301

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q5/5

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q1/5

2303

QUESTION ID

S56-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q1/5

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q2/5

2304

QUESTION ID

S56-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q2/5

How many angles does a quadrilateral have?

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q3/5

2305

QUESTION ID

S56-B1-Q3

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

Every quadrilateral is a square.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q4/5

2306

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q4/5

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q5/5

2307

QUESTION ID

S56-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A trapezoid with exactly one pair of parallel sides

GRADING NOTE Any quadrilateral with fewer than two pairs of parallel sides is correct, for example a trapezoid with exactly one pair of parallel sides or a kite with no parallel sides. A rhombus, rectangle or square is wrong because each is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2309

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2310

QUESTION ID

S56-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2311

QUESTION ID

S56-B2-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

The sides of a square that meet at a corner are perpendicular.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2312

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2313

QUESTION ID

S56-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 4 sides of equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q1/5

2315

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q1/5

A rhombus has a base of 7 in. and a height of 5 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q2/5

2316

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q2/5

A trapezoid has bases of 4 m and 10 m and a height of 6 m. It is cut into a 4-m by 6-m rectangle and two triangles, each with a base of 3 m and a height of 6 m.

What is the area of the trapezoid?

- A.** 84 square meters **B.** 42 square meters
C. 60 square meters **D.** 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q3/5

2317

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 42 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q3/5

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q4/5

2318

QUESTION ID

S56-F1-Q4

TYPE

Short answer

CORRECT ANSWER

32 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q4/5

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q5/5

2319

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 5.G.B.4

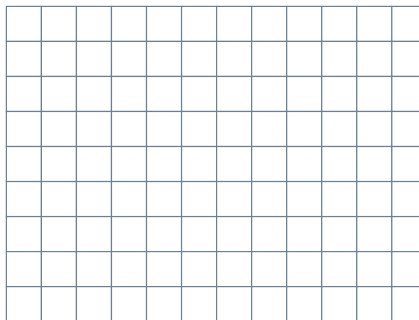
FORWARD 2

Draw quadrilaterals and triangles with given conditions • Grade 7 • Geometry •
7.G.A.2

Draw a rectangle that is not a square.

Write its side lengths.

Side lengths: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q1/5

ID S56-F2-Q1

2321

QUESTION ID

S56-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rectangle drawn along grid lines with two different side lengths

Written: Two different lengths that match the drawing, for example 6 units by 3 units

GRADING NOTE Grade both: four right angles with unequal adjacent sides in the drawing, and side lengths that match it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q1/5

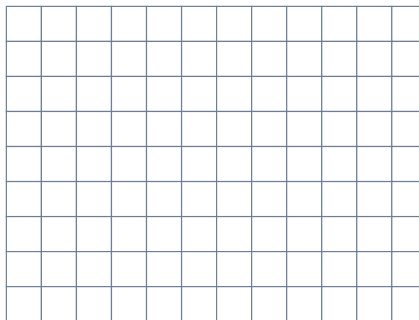
ID S56-F2-Q1

Draw a rhombus that is not a square, with every side 5 units long.

(Hint: a side can go 3 units across and 4 units up.)

How do you know it is not a square?

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q2/5

ID S56-F2-Q2

2322

QUESTION ID

S56-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A quadrilateral with four 5-unit sides whose angles are not right angles, for example with vertices (0, 0), (5, 0), (8, 4) and (3, 4)

Written: Its angles are not right angles.

GRADING NOTE Grade both: four equal sides that are not perpendicular, and a reason that names the angles (not right angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q2/5

ID S56-F2-Q2

Can a quadrilateral have exactly 3 right angles? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q3/5

2323

QUESTION ID

S56-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE The angles of a quadrilateral add to 360° . If three are 90° , the fourth is $360 - 270 = 90^\circ$, so it has 4 right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

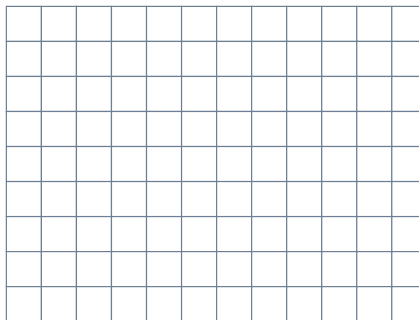
ID S56-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q3/5

Draw a parallelogram that is not a rectangle, with a base of 6 units and a height of 3 units.

What is its area?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q4/5

ID S56-F2-Q4

2324

QUESTION ID

S56-F2-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A parallelogram with a 6-unit base along a grid line and the opposite side 3 units above it, shifted sideways so the angles are not right angles

Written: 18 square units

GRADING NOTE Grade both: the drawing meets all three conditions, and the area 18 square units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q4/5

ID S56-F2-Q4

A triangle has sides of 6 cm, 6 cm, and 6 cm. Can triangles of different shapes be drawn with these sides? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q5/5

2325

QUESTION ID

S56-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Three side lengths determine one unique triangle (here an equilateral triangle); every copy is the same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q5/5

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Right, acute, and obtuse triangles	4.G.A.2
BACKWARD 2	Kinds of angles	4.G.A.1
BACKWARD 3	Triangles	2.G.A.1
FORWARD 1	Area of triangles	6.G.A.1
FORWARD 2	Write and solve equations for unknown angles	7.G.B.5

SET 57 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A.** Obtuse
- B.** Scalene
- C.** Equilateral
- D.** Right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q1/5

2328

QUESTION ID

S57-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Equilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q1/5

A triangle has angles of 90° , 50° , and 40° .

Classify it by its angles.

Classification: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q2/5

2329

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q2/5

True or false?

An isosceles triangle has at least two sides of equal length. Every equilateral triangle is also an isosceles triangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q3/5

2330

QUESTION ID

S57-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q3/5

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

- A.** Acute equilateral
- B.** Right scalene
- C.** Right isosceles
- D.** Obtuse scalene

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q4/5

2331

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Obtuse scalene

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q4/5

A triangle has angles of 70° , 70° , and 40° .

Classify it by its angles and by its sides.

By angles: _____

By sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q5/5

2332

QUESTION ID

S57-M-Q5

TYPE

Short answer

CORRECT ANSWER

Acute; isosceles

GRADING NOTE Two equal angles mean two equal sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q5/5

SET 57 • 5.G.B.4

BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q1/5

2334

QUESTION ID

S57-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q1/5

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q2/5

2335

QUESTION ID

S57-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q2/5

A triangle has one angle greater than 90° .

What kind of triangle is it?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q3/5

2336

QUESTION ID

S57-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q3/5

True or false?

In an acute triangle, all three angles are less than 90° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q4/5

2337

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q4/5

Which angle measure is acute?

- A.** 100°
- B.** 180°
- C.** 60°
- D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q5/5

2338

QUESTION ID

S57-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 5.G.B.4

BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q1/5

2340

QUESTION ID

S57-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q1/5

What kind of angle measures 35° ?

- A. Right
- B. Straight
- C. Obtuse
- D. Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q2/5

2341

QUESTION ID

S57-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q2/5

What kind of angle measures 150° ?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q3/5

2342

QUESTION ID

S57-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q3/5

True or false?

Two right angles together make 360° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q4/5

2343

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q4/5

What kind of angle measures exactly 90° ?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q5/5

2344

QUESTION ID

S57-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 5.G.B.4

BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q1/5

2346

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q1/5

True or false?

A triangle has 3 angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q2/5

2347

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q2/5

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q3/5

2348

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q3/5

True or false?

All triangles have sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q4/5

2349

QUESTION ID

S57-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q4/5

How many vertices (corners) does a triangle have?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q5/5

2350

QUESTION ID

S57-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q1/5

2352

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q1/5

A triangle has a base of 10 in. and a height of 7 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q2/5

2353

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q2/5

An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters **B.** 40 square meters
C. 10 square meters **D.** 9 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q3/5

2354

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 square meters

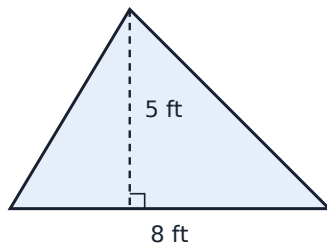
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q3/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q4/5

2355

QUESTION ID

S57-F1-Q4

TYPE

Short answer

CORRECT ANSWER

20 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q4/5

True or false?

Two triangles with the same base and the same height have the same area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q5/5

2356

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 5.G.B.4

FORWARD 2

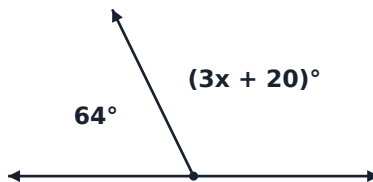
Write and solve equations for unknown angles • Grade 7 • Geometry • 7.G.B.5

The two angles form a straight line.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q1/5

2358

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$(3x + 20) + 64 = 180; x = 32$$

GRADING NOTE Any equivalent equation is correct (for example $3x + 84 = 180$). Both the equation and $x = 32$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

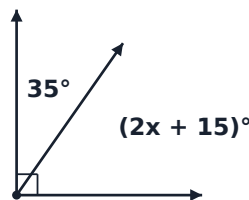
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q1/5

The two angles together form a right angle.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q2/5

2359

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$$(2x + 15) + 35 = 90;$$
$$x = 20$$

GRADING NOTE Any equivalent equation is correct (for example $2x + 50 = 90$). Both the equation and $x = 20$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

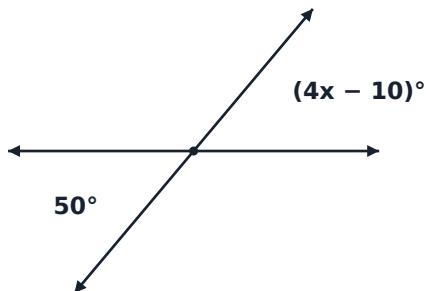
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q2/5

Two lines cross. The marked angles are vertical angles.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q3/5

ID S57-F2-Q3

2360

QUESTION ID

S57-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$4x - 10 = 50; x = 15$

GRADING NOTE Vertical angles are equal. Any equivalent equation is correct. Both the equation and $x = 15$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

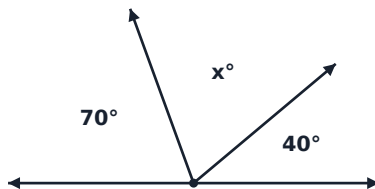
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q3/5

ID S57-F2-Q3

Three angles together form a straight line.

Which equation can be used to find x ?

- A. $x = 40 \times 70$
- B. $40 + x + 70 = 180$
- C. $40 + x = 70$
- D. $40 + x + 70 = 90$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q4/5

2361

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 + x + 70 = 180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q4/5

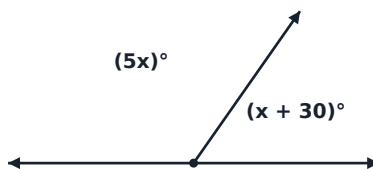
Angle ABC and angle CBD together form a straight line. Angle ABC is $(5x)^\circ$ and angle CBD is $(x + 30)^\circ$.

Write an equation, solve for x , and find both angles.

Equation: _____

$x =$ _____

Angles: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q5/5

ID S57-F2-Q5

2362

QUESTION ID

S57-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**$5x + (x + 30) = 180;$
 $x = 25;$ angle
ABC = 125° , angle
CBD = 55°**

GRADING NOTE Any equivalent equation is correct ($6x + 30 = 180$). All three parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q5/5

ID S57-F2-Q5

Place triangles in a hierarchy of categories and subcategories

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Classify triangles by their angles	4.G.A.2
BACKWARD 2	Shapes in different categories share attributes	3.G.A.1
FORWARD 1	Area of special triangles	6.G.A.1
FORWARD 2	Decide whether conditions give one triangle, more than one, or no...	7.G.A.2

SET 71 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A. Triangle, isosceles triangle, equilateral triangle
- B. Isosceles triangle, triangle, equilateral triangle
- C. Triangle, equilateral triangle, isosceles triangle
- D. Equilateral triangle, isosceles triangle, triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q1/5

2365

QUESTION ID

S71-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Triangle, isosceles triangle, equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q1/5

True or false?

Every right isosceles triangle belongs to both the category of right triangles and the category of isosceles triangles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q2/5

2366

QUESTION ID

S71-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q2/5

Isosceles means at least two sides of equal length.

A triangle has sides of 7, 7, and 7 units. Which of these categories does it belong to?

triangle, isosceles, equilateral, scalene

Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q3/5

2367

QUESTION ID

S71-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Triangle, isosceles,
and equilateral**

GRADING NOTE All three are required. Scalene is wrong because a scalene triangle has no sides of equal length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q3/5

2367

Isosceles means at least two sides of equal length.

Which statement is true?

- A. Some isosceles triangles are equilateral.
- B. No right triangle is isosceles.
- C. Every isosceles triangle is equilateral.
- D. Some scalene triangles are isosceles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q4/5

2368

QUESTION ID

S71-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Some isosceles triangles are equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q4/5

Explain why a triangle cannot be both a right triangle and an obtuse triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q5/5

2369

QUESTION ID

S71-M-Q5

TYPE

Short answer

CORRECT ANSWER

A right triangle has a 90° angle and an obtuse triangle has an angle greater than 90° . The three angles of a triangle add to 180° , so these two angles together would already be more than 180° .

GRADING NOTE Accept the 180° angle-sum reason, or that a triangle can have only one angle of 90° or more because the two sides would never meet. The categories "right" and "obtuse" do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q5/5

SET 71 • 5.G.B.4

BACKWARD 1

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A triangle with angles of 30° , 60° , and 90° is a right triangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q1/5

2371

QUESTION ID

S71-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q1/5

A triangle has angles of 110° , 40° , and 30° .

What kind of triangle is it?

- A. Equilateral
- B. Acute
- C. Right
- D. Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q2/5

2372

QUESTION ID

S71-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q2/5

A triangle has angles of 50° , 60° , and 70° .

Is it acute, right, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q3/5

2373

QUESTION ID

S71-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q3/5

True or false?

A triangle can have two obtuse angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q4/5

2374

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q4/5

Which three angles could be the angles of an acute triangle?

- A. 100° , 50° , 30°
- B. 120° , 30° , 30°
- C. 80° , 60° , 40°
- D. 90° , 45° , 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q5/5

2375

QUESTION ID

S71-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 80° , 60° , 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • 5.G.B.4

BACKWARD 2

Shapes in different categories share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

Rhombuses, rectangles, and squares are all quadrilaterals.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q1/5

2377

QUESTION ID

S71-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q1/5

What do all quadrilaterals have in common?

- A.** 4 right angles
- B.** 4 sides
- C.** 2 pairs of parallel sides
- D.** 4 sides of equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q2/5

2378

QUESTION ID

S71-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 4 sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q2/5

Name a quadrilateral that is not a rectangle.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q3/5

2379

QUESTION ID

S71-B2-Q3

TYPE

Short answer

CORRECT ANSWER

**A rhombus that is
not a square**

GRADING NOTE Any quadrilateral that is not a rectangle is correct, for example a trapezoid or a rhombus that is not a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Every quadrilateral is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q4/5

2380

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q4/5

A square is also a ____.

- A.** rectangle
- B.** pentagon
- C.** hexagon
- D.** triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q5/5

2381

QUESTION ID

S71-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • 5.G.B.4

FORWARD 1

Area of special triangles • Grade 6 • Geometry • 6.G.A.1

A right isosceles triangle has legs that are 6 cm long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q1/5

2383

QUESTION ID

S71-F1-Q1

TYPE

Short answer

CORRECT ANSWER

18 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q1/5

True or false?

A triangle with a base of 4 in. and a height of $3\frac{1}{2}$ in. has an area of 7 square inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q2/5

2384

QUESTION ID

S71-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q2/5

A square with sides of 8 m is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 64 square meters
- B.** 8 square meters
- C.** 32 square meters
- D.** 16 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q3/5

2385

QUESTION ID

S71-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 32 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q3/5

An isosceles triangle has a base of 10 ft and a height of 12 ft.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q4/5

2386

QUESTION ID

S71-F1-Q4

TYPE

Short answer

CORRECT ANSWER

60 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 5 units and 9 units has an area of 45 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q5/5

2387

QUESTION ID

S71-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q5/5

SET 71 • 5.G.B.4

FORWARD 2

Decide whether conditions give one triangle, more than one, or none • Grade 7
• Geometry • 7.G.A.2

How many differently shaped triangles can be drawn with sides of 5 cm, 5 cm, and 8 cm?

- A. Exactly two
- B. More than one
- C. Exactly one
- D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q1/5

2389

QUESTION ID

S71-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Exactly one

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q1/5

True or false?

Triangles of many different sizes can be drawn with angles of 60° , 60° , and 60° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q2/5

2390

QUESTION ID

S71-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q2/5

Can a triangle have sides of 3 m, 4 m, and 8 m? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q3/5

2391

QUESTION ID

S71-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE $3 + 4 = 7$ is less than 8, so the two shorter sides cannot meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q3/5

True or false?

A triangle can have angles of 90° , 60° , and 40° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q4/5

2392

QUESTION ID

S71-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q4/5

Can a right triangle also be isosceles? If it can, give its angle measures.

Answer: _____

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q5/5

2393

QUESTION ID

S71-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes; 90°, 45°, 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q5/5